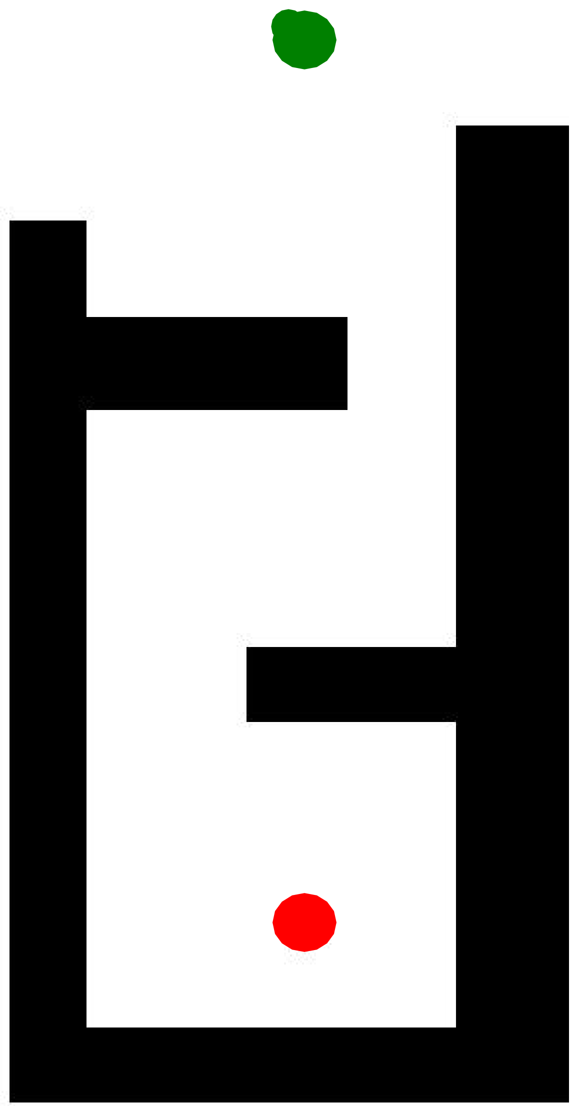
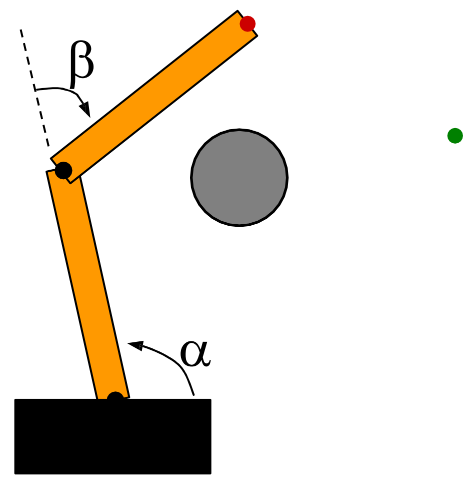
Motion Planning: Configuration Space

What if the robot is not a point?

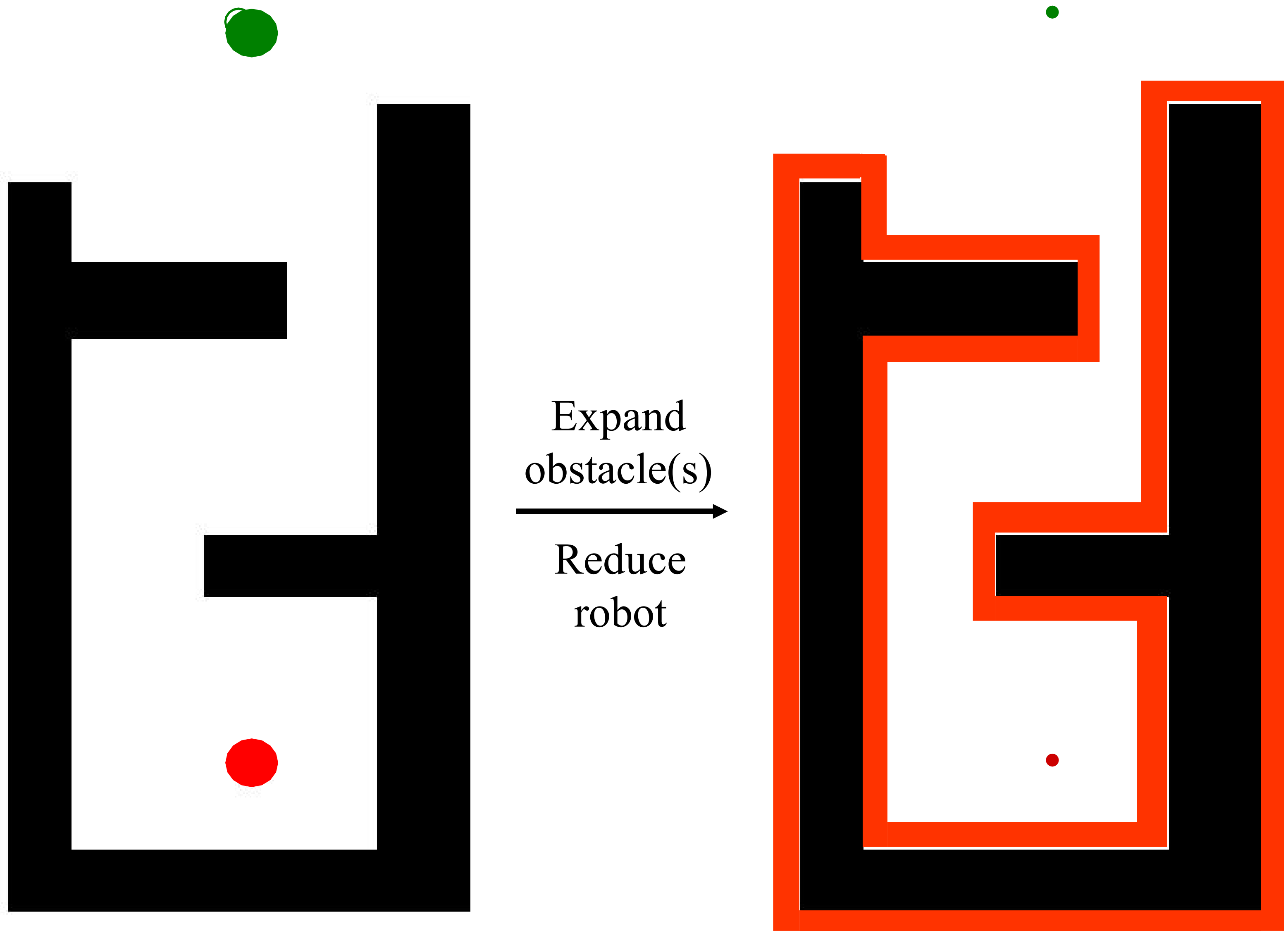


The Scout should probably not be modeled as a point...



Nor should robots with extended linkages that may contact obstacles...

What is the position of the robot?



Configuration Space

- A key concept for motion planning is a **configuration**:
  - *a complete specification of the position of every point in the system*

- A simple example: a robot that translates but does not rotate in the plane:

  - what is a sufficient representation of its configuration?

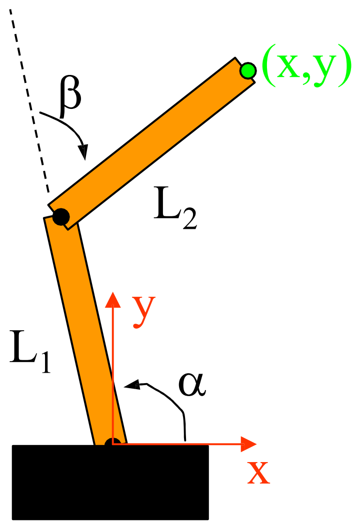
- The space of all configurations is the **configuration space** or **C-space.**

- C-space formalism: Lozano-Perez ‘79

Robot Manipulators

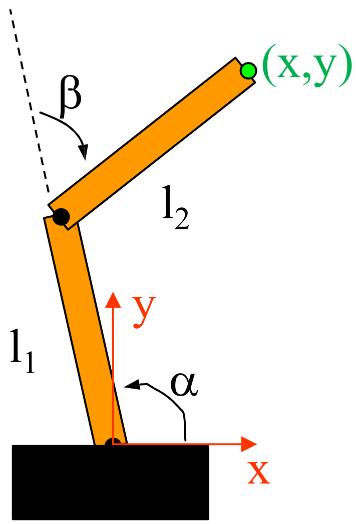
What are this arm's forward kinematics?

(How does its position depend on its joint angles?)



Robot Manipulators

What are this arm's forward kinematics?



Find **(x,y)** in terms of α and β ...

Keeping it "simple"

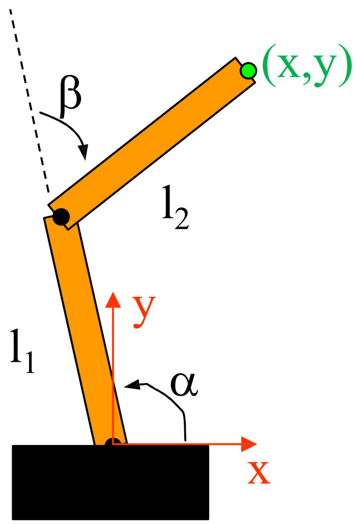
$$c_\alpha = \cos(\alpha),\ s_\alpha = \sin(\alpha),$$

$$c_\beta = \cos(\beta)\,,\ s_\beta = \sin(\beta)$$

$$c_+ = \cos(\alpha + \beta),$$

$$s_+ = \sin(\alpha + \beta)$$

Manipulator kinematics



$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} l_1 c_\alpha \\ l_1 s_\alpha \end{pmatrix} + \begin{pmatrix} l_2 c_+ \\ l_2 s_+ \end{pmatrix}$$ Position

Keeping it "simple"

$c_\alpha = \cos(\alpha), s_\alpha = \sin(\alpha),$

$c_\beta = \cos(\beta), s_\beta = \sin(\beta)$

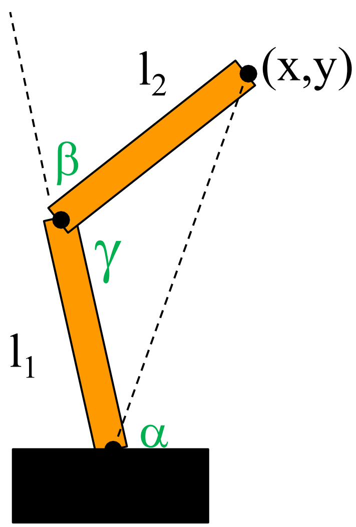
$c_+ = \cos(\alpha + \beta),$

$s_+ = \sin(\alpha + \beta)$

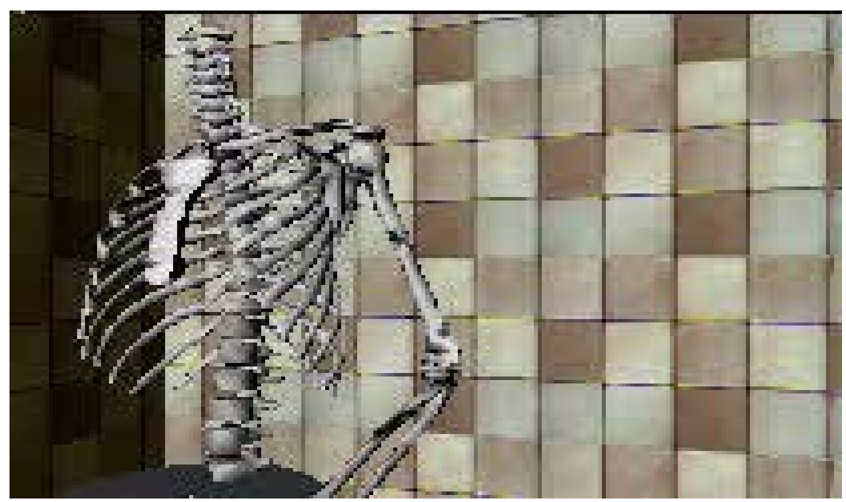
Inverse Kinematics

Inverse kinematics -- finding joint angles from Cartesian coordinates

via a geometric or algebraic approach...



Given (x,y) and l_1 and l_2, what are the values of α, β, γ?



Inverse Kinematics

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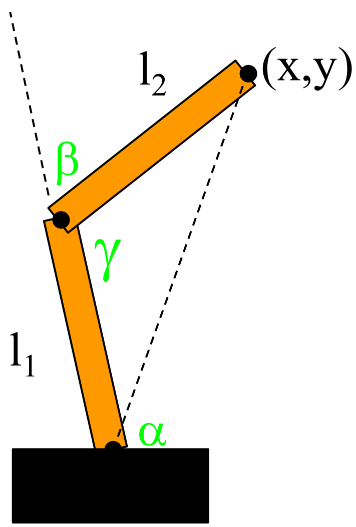
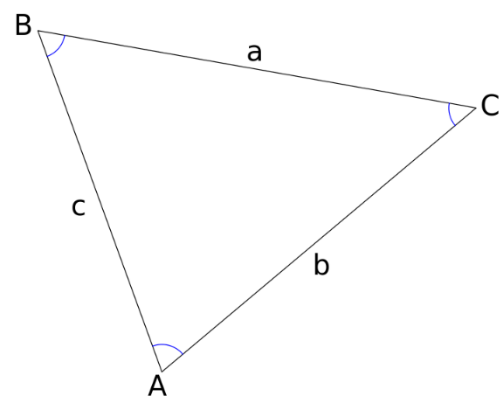




$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$c^2 = a^2 + b^2 \ - 2ab \cos C$$

$$\gamma = \cos^{-1}\left(\frac{x^2 + y^2 - l_1^2 - l_2^2}{-2 l_1 l_2}\right) \qquad \beta = 180 \ - \gamma$$

$$\alpha = \sin^{-1}\left(\frac{l_2 \sin(\gamma)}{\sqrt{x^2 + y^2}}\right) + \tan^{-1}\left(\frac{y}{x}\right)$$

atan2(y,x)

Puma Inv. Kinematics

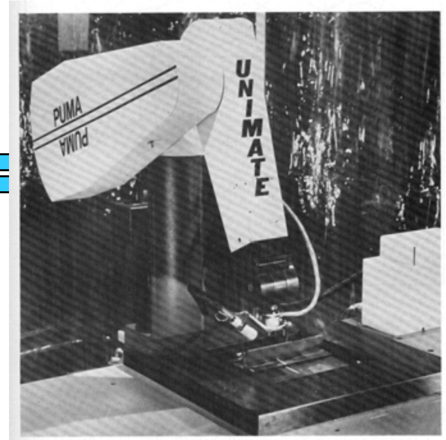


FIGURE 3.17 The Unimation PUMA 560.

```
%
% Solve for theta(1)

r=sqrt(Px^2 + Py^2);
if (n1 == 1),
    theta(1)= atan2(Py,Px) + asin(d3/r);
else
    theta(1)= atan2(Py,Px) + pi - asin(d3/r);
end

%
% Solve for theta(2)

V114= Px*cos(theta(1)) + Py*sin(theta(1));
r=sqrt(V114^2 + Pz^2);
Psi = acos((a2^2-d4^2-a3^2+V114^2+Pz^2)/
           (2.0*a2*r));
theta(2) = atan2(Pz,V114) + n2*Psi;

%
% Solve for theta(3)

num = cos(theta(2))*V114+sin(theta(2))*Pz-a2;
den = cos(theta(2))*Pz - sin(theta(2))*V114;
theta(3) = atan2(a3,d4) - atan2(num, den);
```

```
% Solve for theta(4)

V113 = cos(theta(1))*Ax + sin(theta(1))*Ay;
V323 = cos(theta(1))*Ay - sin(theta(1))*Ax;
V313 = cos(theta(2)+theta(3))*V113 +
       sin(theta(2)+theta(3))*Az;
theta(4) = atan2((n4*V323),(n4*V313));

% Solve for theta(5)

num = -cos(theta(4))*V313 - V323*sin(theta(4));
den = -V113*sin(theta(2)+theta(3)) +
       Az*cos(theta(2)+theta(3));
theta(5) = atan2(num,den);

% Solve for theta(6)

V112 = cos(theta(1))*Ox + sin(theta(1))*Oy;
V132 = sin(theta(1))*Ox - cos(theta(1))*Oy;
V312 = V112*cos(theta(2)+theta(3)) +
       Oz*sin(theta(2)+theta(3));
V332 = -V112*sin(theta(2)+theta(3)) +
        Oz*cos(theta(2)+theta(3));
V412 = V312*cos(theta(4)) - V132*sin(theta(4));
V432 = V312*sin(theta(4)) + V132*cos(theta(4));
num = -V412*cos(theta(5)) - V332*sin(theta(5));
den = - V432;
theta(6) = atan2(num,den);
```

Some Other Examples of C-Space

- A rotating bar fixed at a point
  - what is its C-space?
  - what is its workspace
- A rotating bar that translates along the rotation axis
  - what is its C-space?
  - what is its workspace
- A two-link manipulator
  - what is its C-space?
  - what is its workspace?
  - Suppose there are joint limits, does this change the C-space?
  - The workspace?

Obstacles in C-Space

- Let q denote a point in a configuration space Q
- The path planning problem is to find a mapping $c:[0,1] \rightarrow Q$ s.t. no configuration along the path intersects an obstacle
- Recall a workspace obstacle is WO_i
- A *configuration space obstacle* QO_i is the set of configurations q at which the robot intersects WO_i, *that is*

$$QO_i = \left\{ q \in Q \mid R(q) \bigcap WO_i \neq 0 \right\}$$

- The *free configuration space (*or just *free space*) Q_{free} *is*

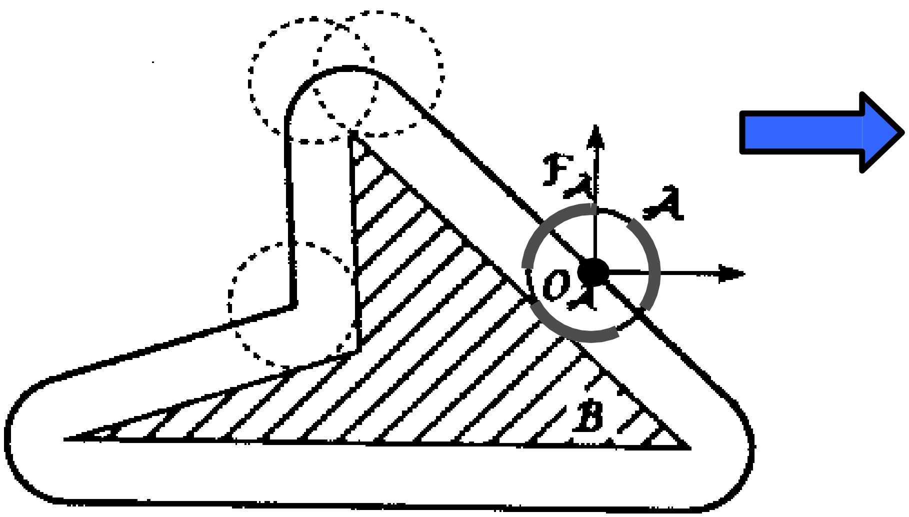
$$\mathrm{Q}_{\mathrm{free}} = \mathrm{Q} \setminus (\cup QO_i)$$

The free space is generally an open set A *free path* is a mapping $c:[0,1] \rightarrow Q_{free}$

A *semifree path* is a mapping $c:[0,1] \rightarrow \mathrm{cl}(Q_{free})$

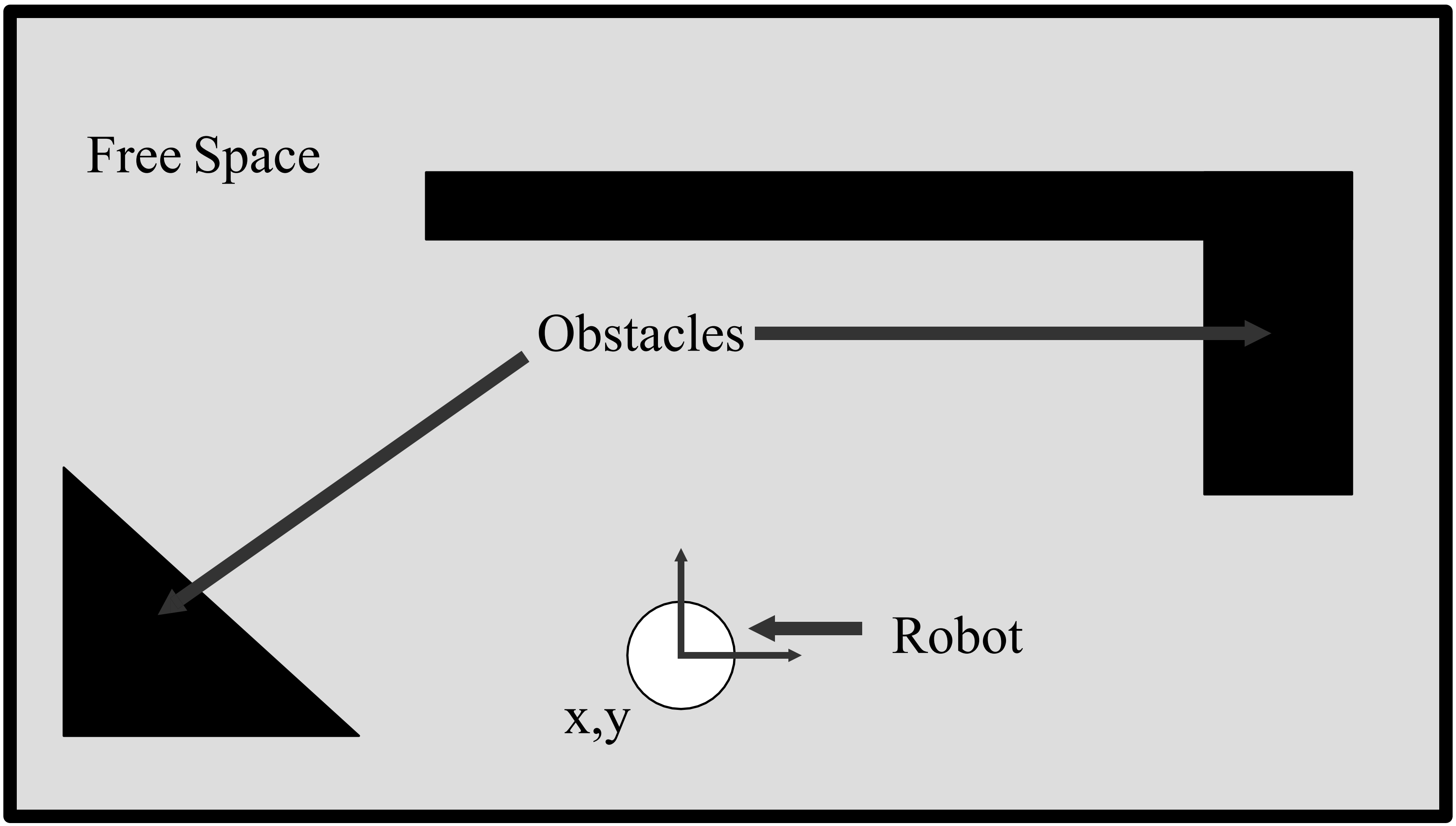
Disc in 2-D workspace

workspace
workspace

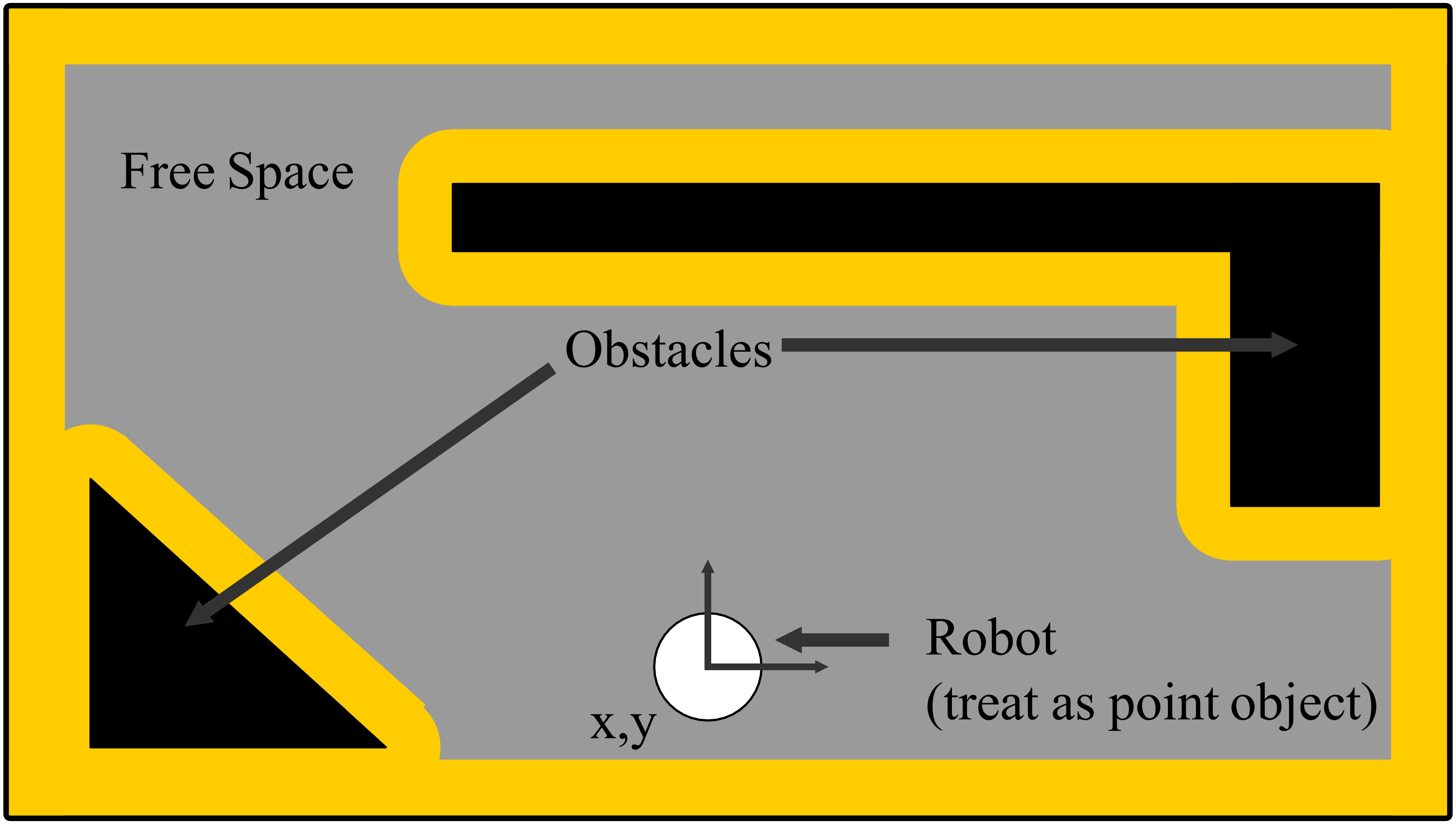


configuration space

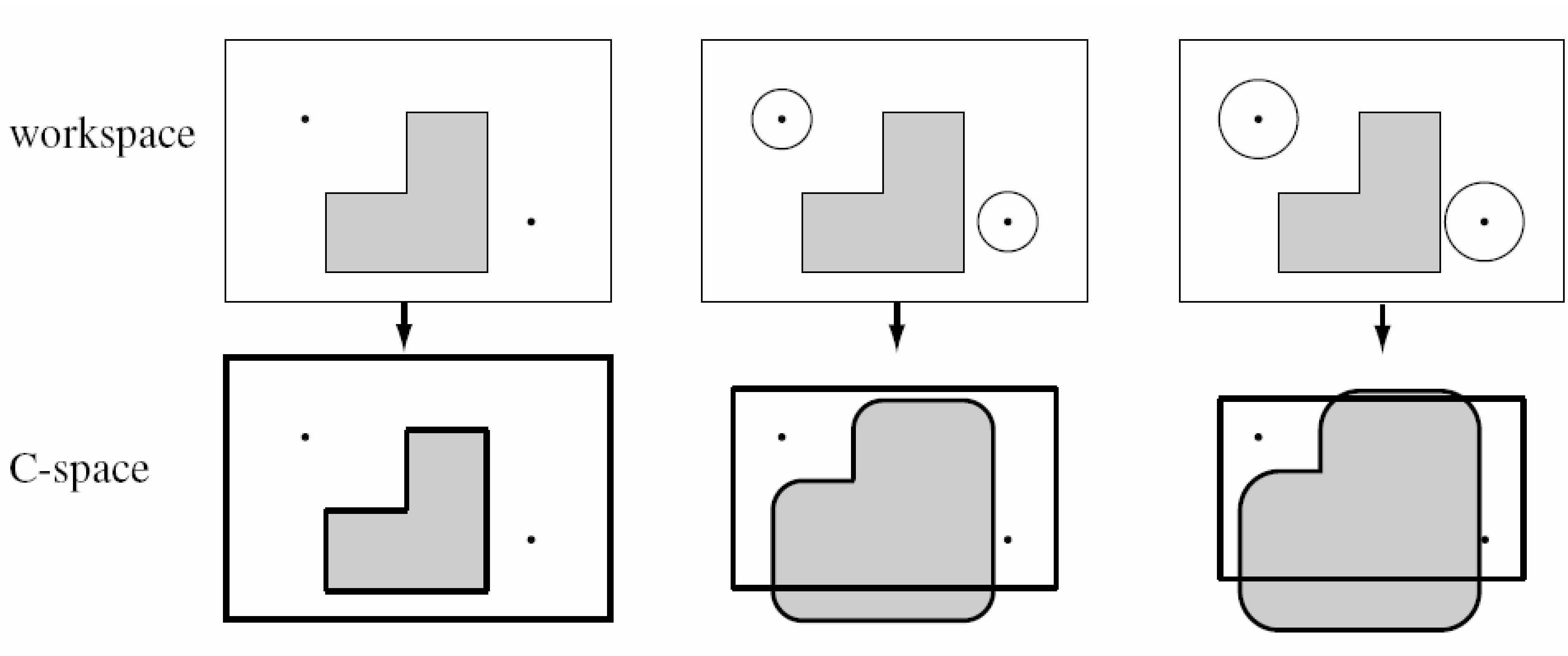
Example of a World (and Robot)



Configuration Space: Accommodate Robot Size



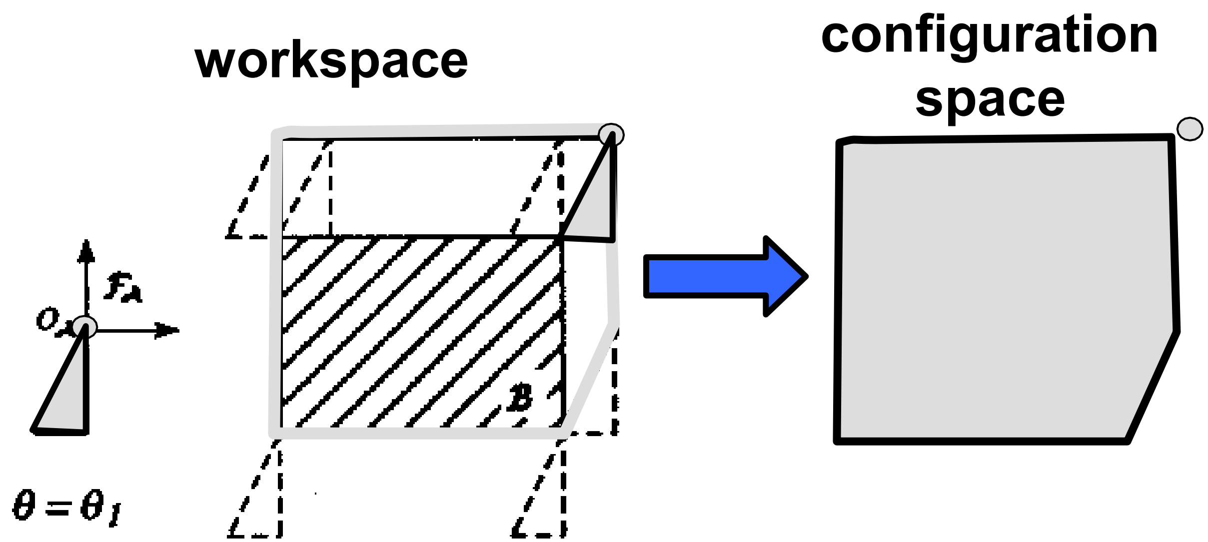
Trace Boundary of Workspace



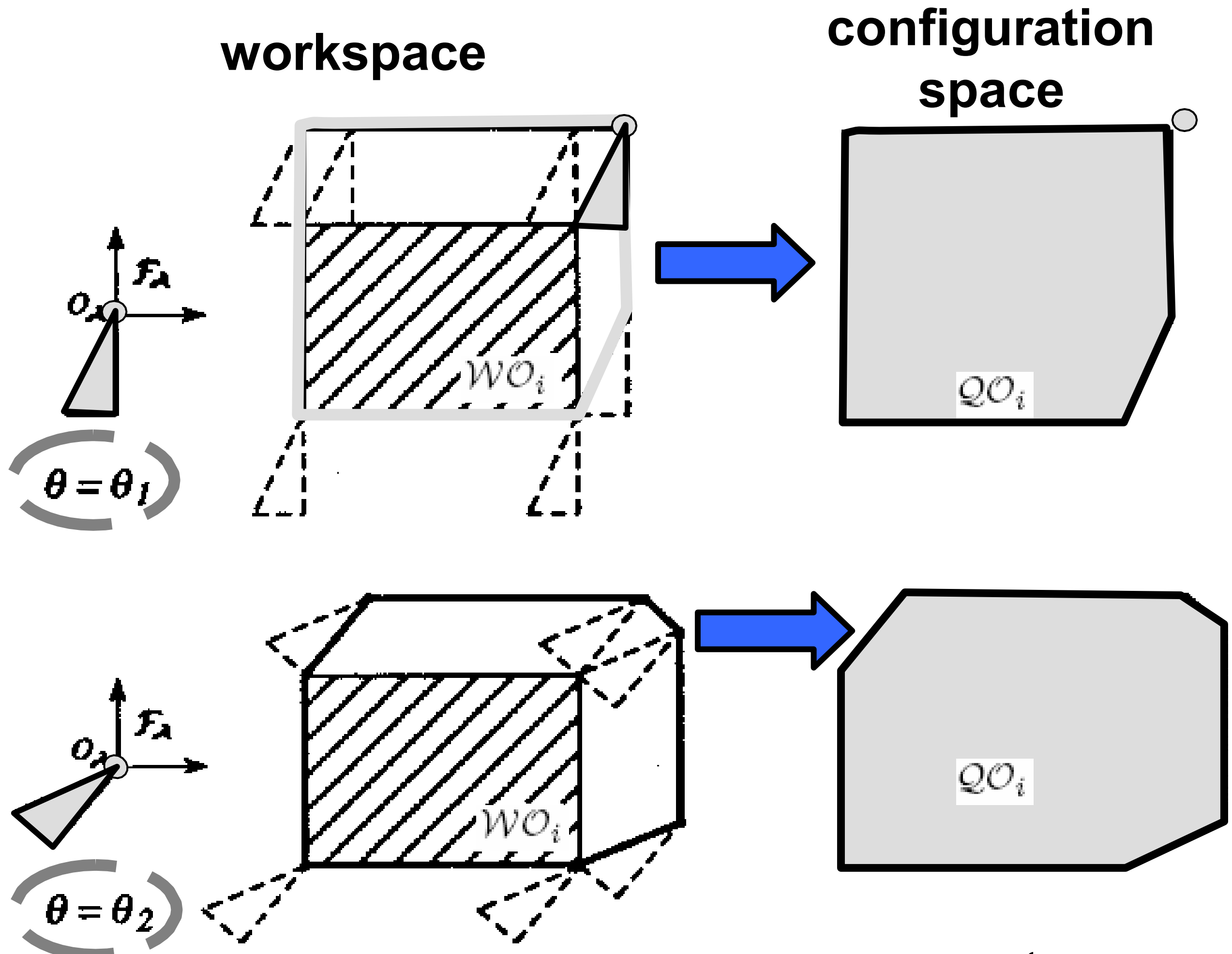
$$QO_i = \left\{ q \in Q \mid R(q) \bigcap WO_i \neq 0 \right\}$$

Pick a reference point…

Polygonal robot translating in 2-D workspace

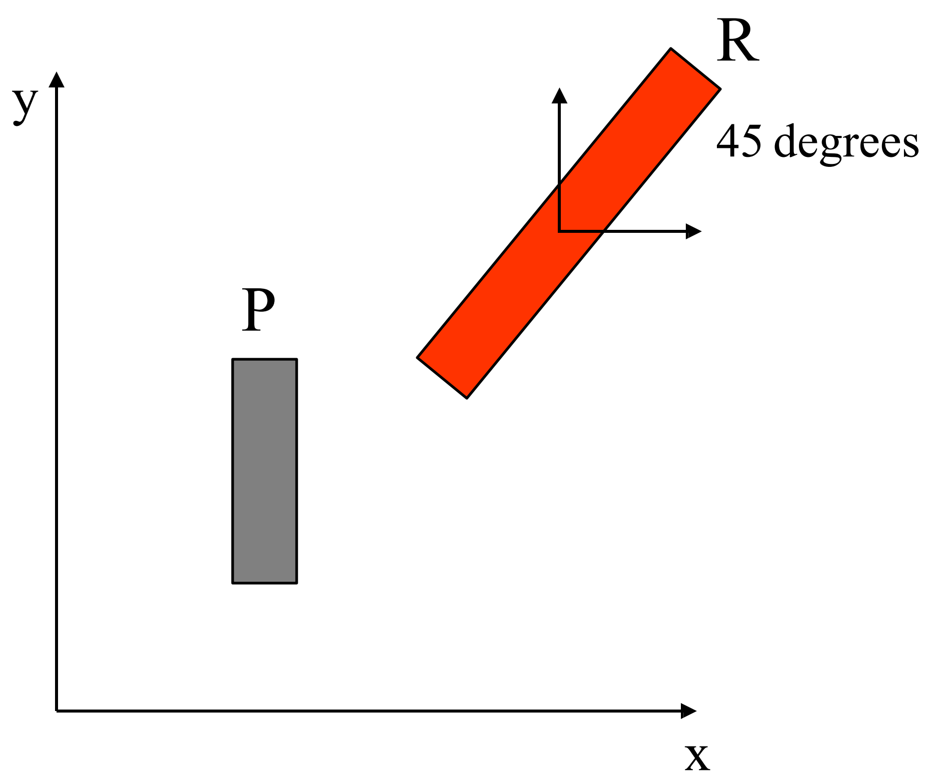


Polygonal robot translating & rotating in 2-D workspace



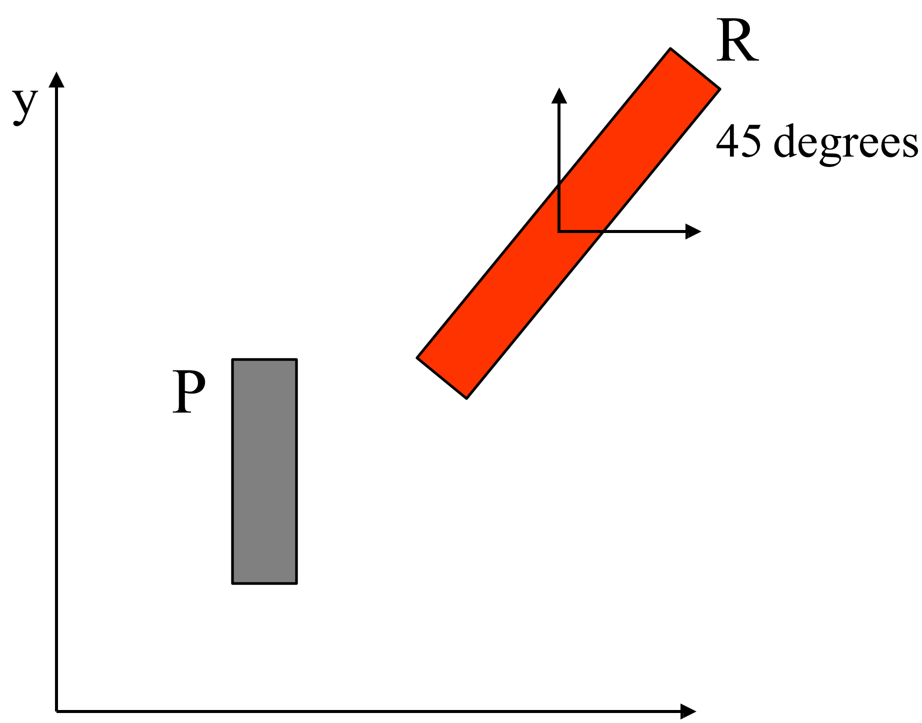
$$QO_i = \left\{ q \in Q \mid R(q) \bigcap WO_i \neq 0 \right\}$$

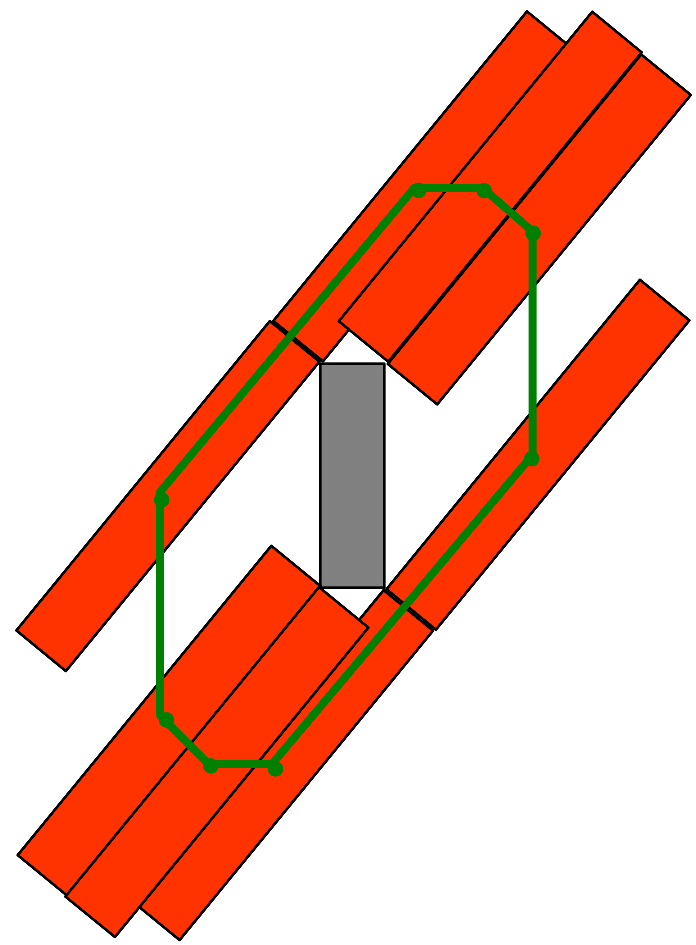
Any reference point



Any reference point configuration

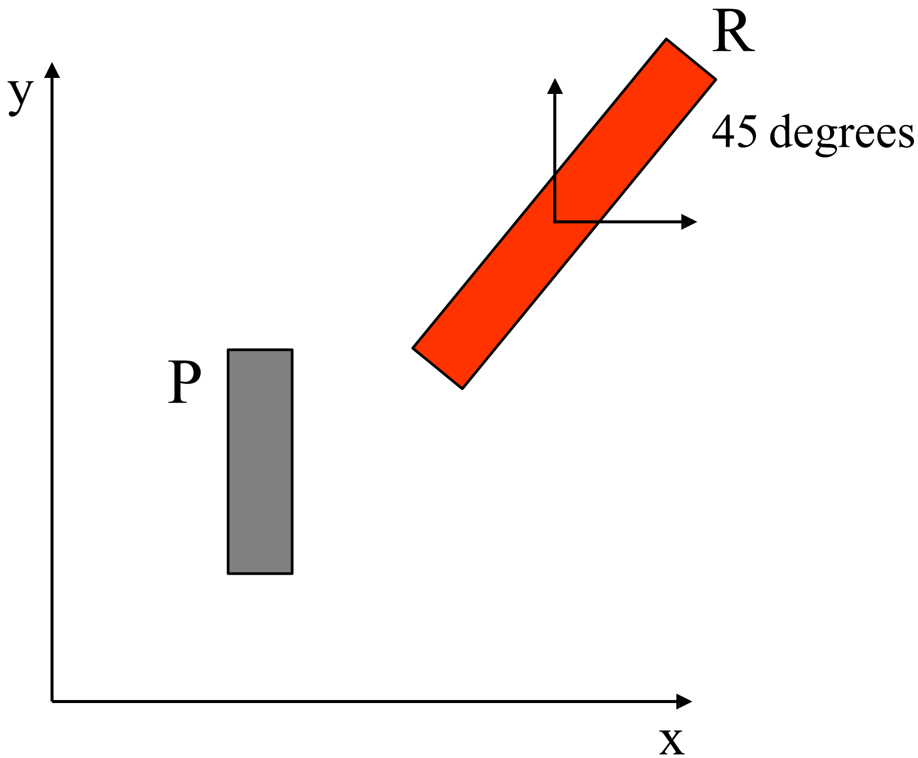
Taking the cross section of configuration space in which the robot is rotated 45 degrees...

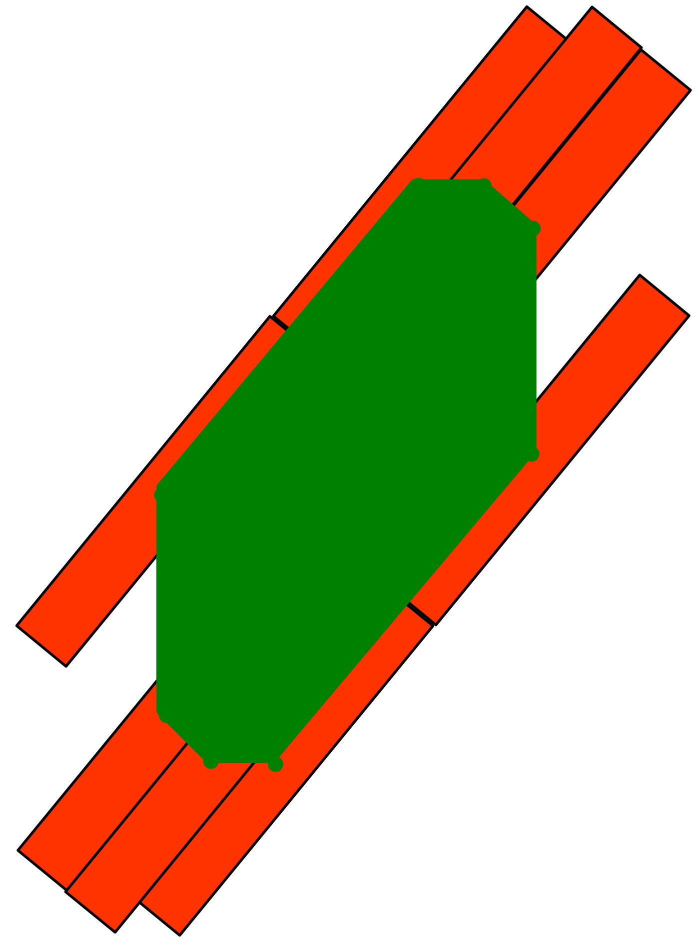




Any reference ~~point~~ configuration

Taking the cross section of configuration space
in which the robot is rotated 45 degrees...

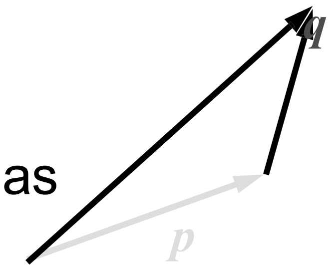




Minkowski sum

- The **Minkowski sum** of two sets P and Q, denoted by $P \oplus Q$, is defined as

$$P \oplus Q = \{p + q \mid p \in P, \quad q \in Q\}$$



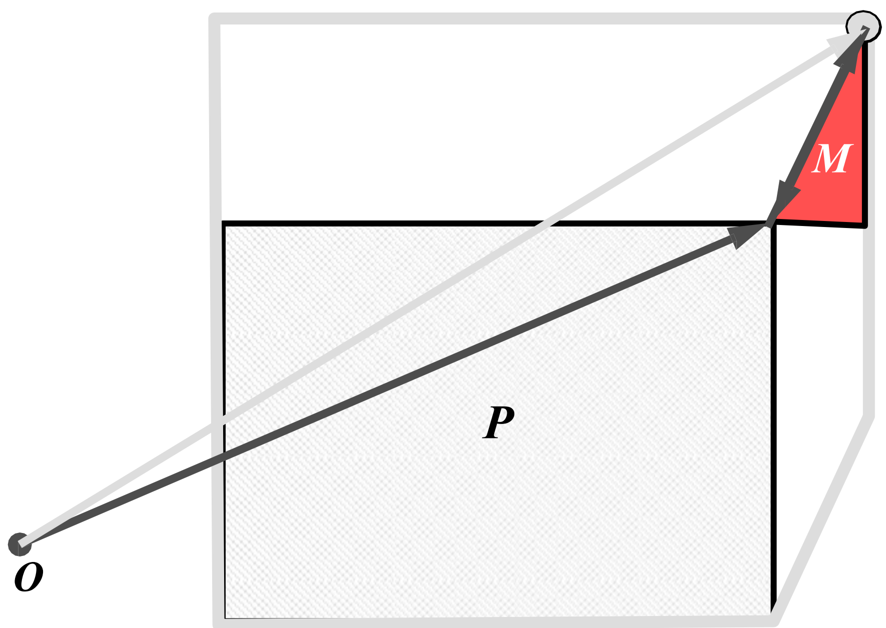
- Similarly, the **Minkowski difference** is defined as

$$P \ominus Q = \{p - q \mid p \in P, \quad q \in Q\}$$

- The Minkowski sum of two **convex** polygons P and Q of m and n vertices respectively is a convex polygon $P \oplus Q$ of $m + n$ vertices.
  - The vertices of $P \oplus Q$ are the "sums" of vertices of P and Q.

Observation

- If P is an obstacle in the workspace and M is a moving object. Then the C-space obstacle corresponding to P is $P \ominus M(0)$



Star Algorithm: Polygonal Obstacles

The method is based on

- sorting normals to the edges of the polygons on the basis of angles.

The **key observation** is that every edge of C_{obs} is a translated edge from either R(obot) or O(bstacle).
In fact, every edge from O and R is used exactly once in the construction of C_{obs}. The only problem is to determine the ordering of these edges of C_{obs}.

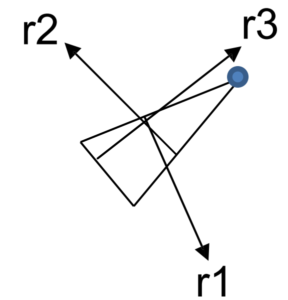
- Let 1, 2, . . ., n denote the angles of the inward edge normal in counterclockwise order around R.
- Let 1, 2, . . ., n denote the outward edge normals to O.

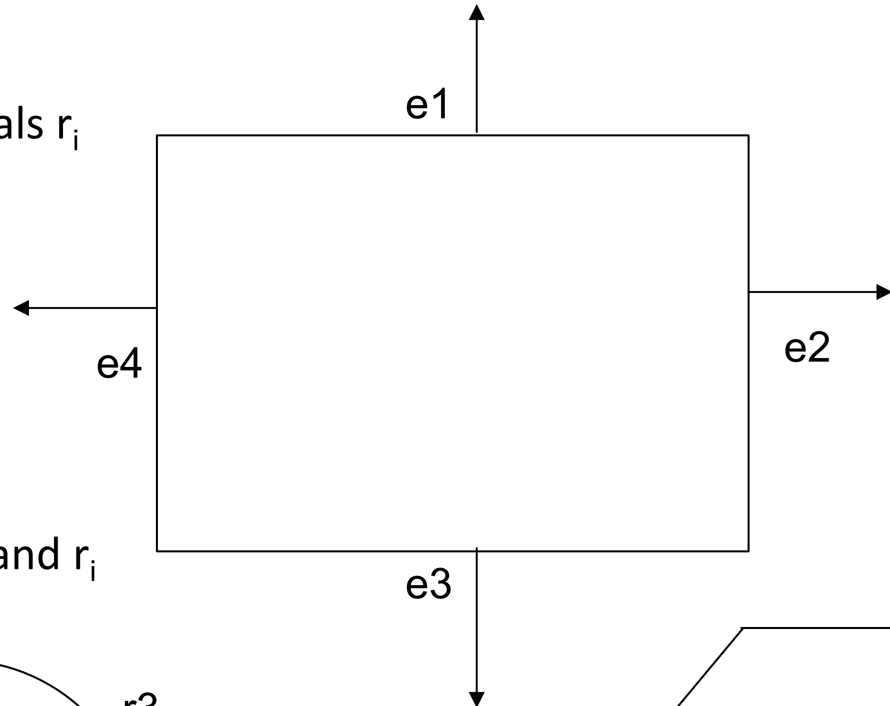
After sorting both sets of angles in circular order around S_1, C_{obs} can be constructed incrementally by using the edges that correspond to the sorted normals, in the order in which they are encountered.

This gives the order of edges in C_{obs}.

Star Algorithm: Polygonal Obstacles

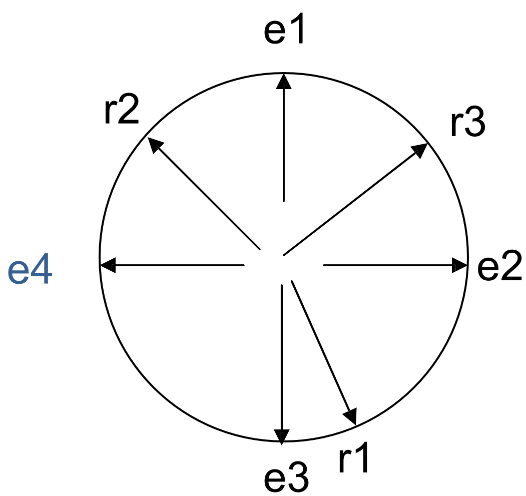
inbound normals r_i

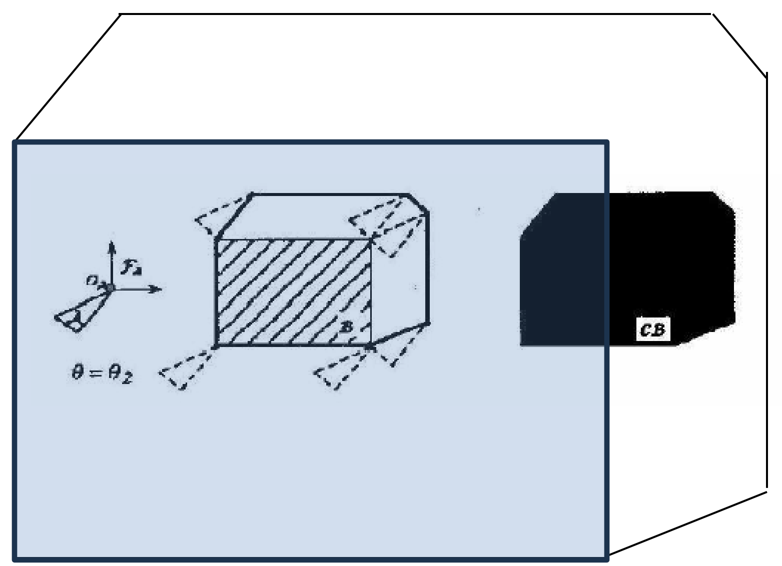




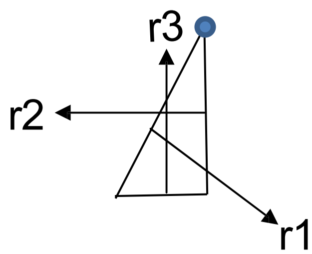
outbound normals e_i

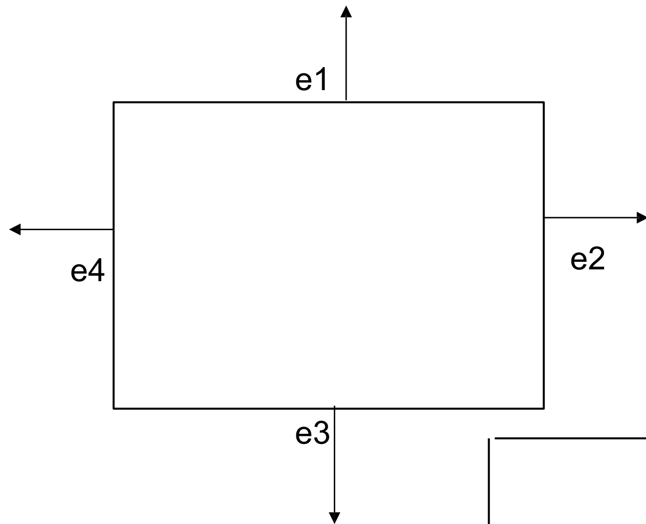
sorted normals e_i and r_i

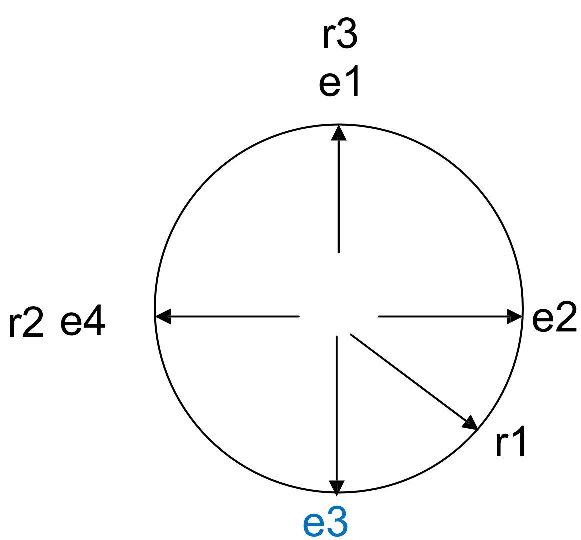


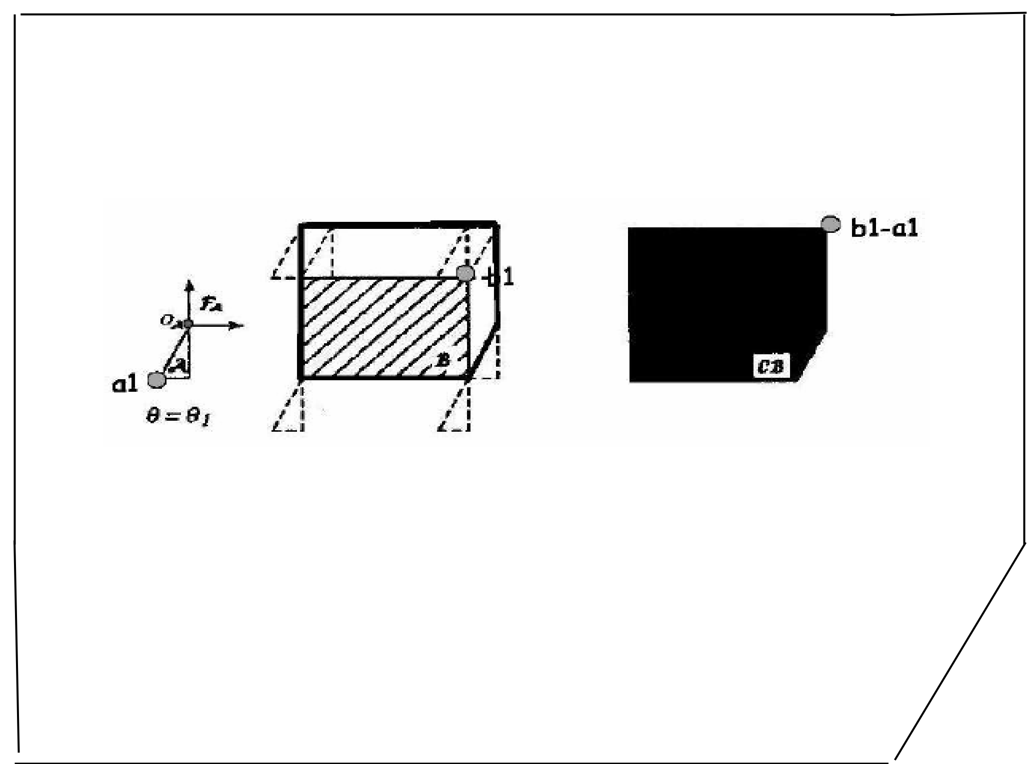


Star Algorithm: 2nd example

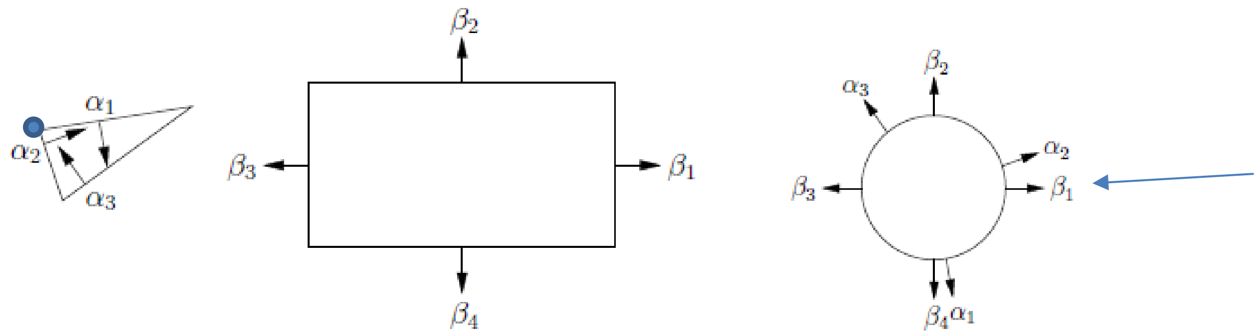




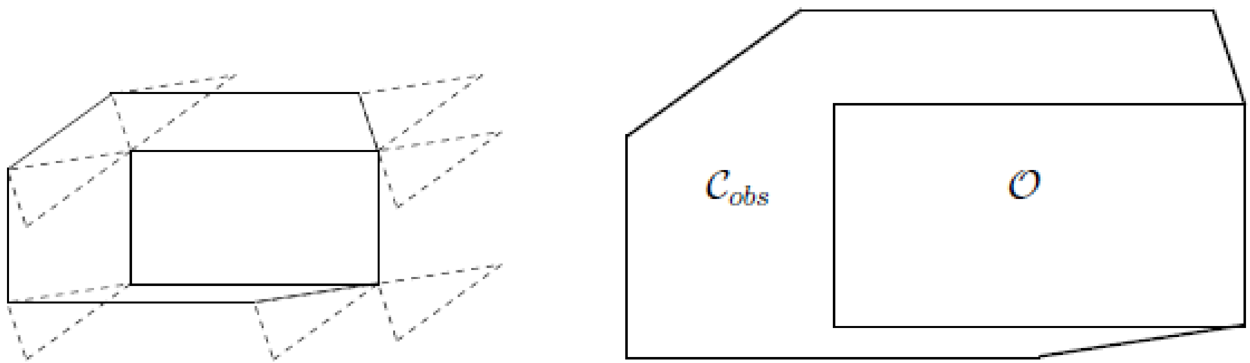




Star Algorithm: 3rd example



(a) (b)

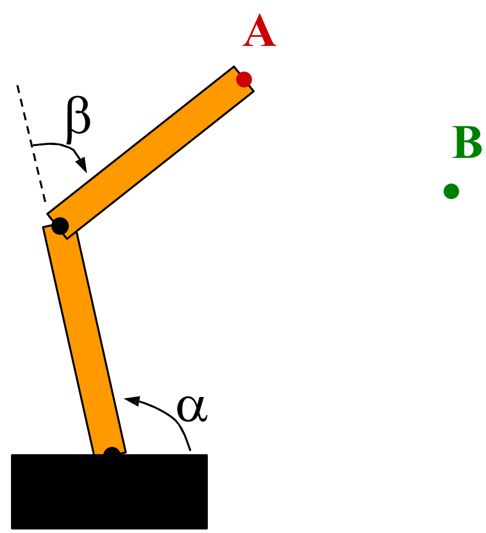


(a) (b)

Start Point

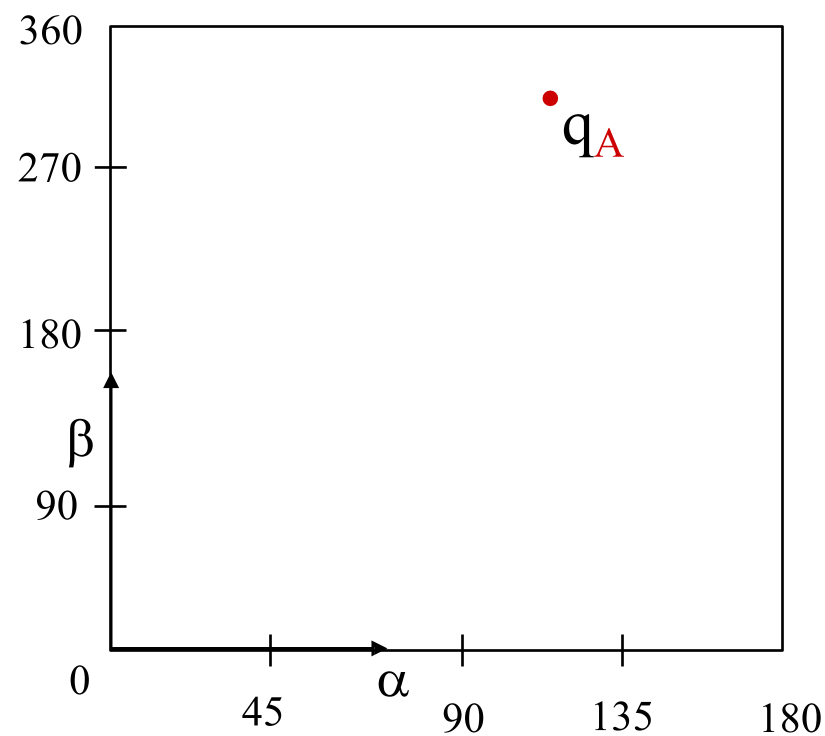
- Leave that as an exercise for your homework.

Configuration Space



An obstacle in the robot's workspace

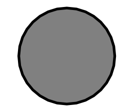
Where can we put $\bullet\, q_B$?

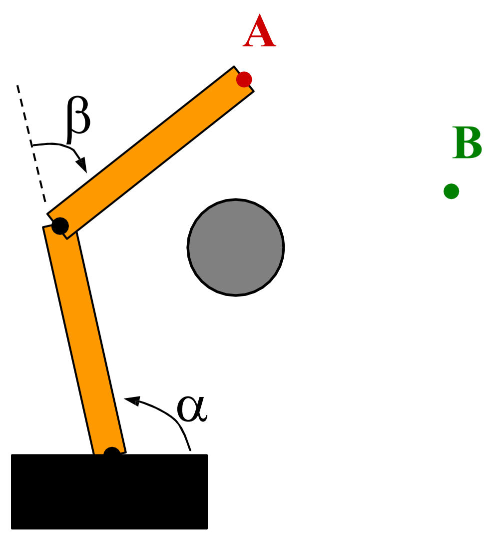


Torus

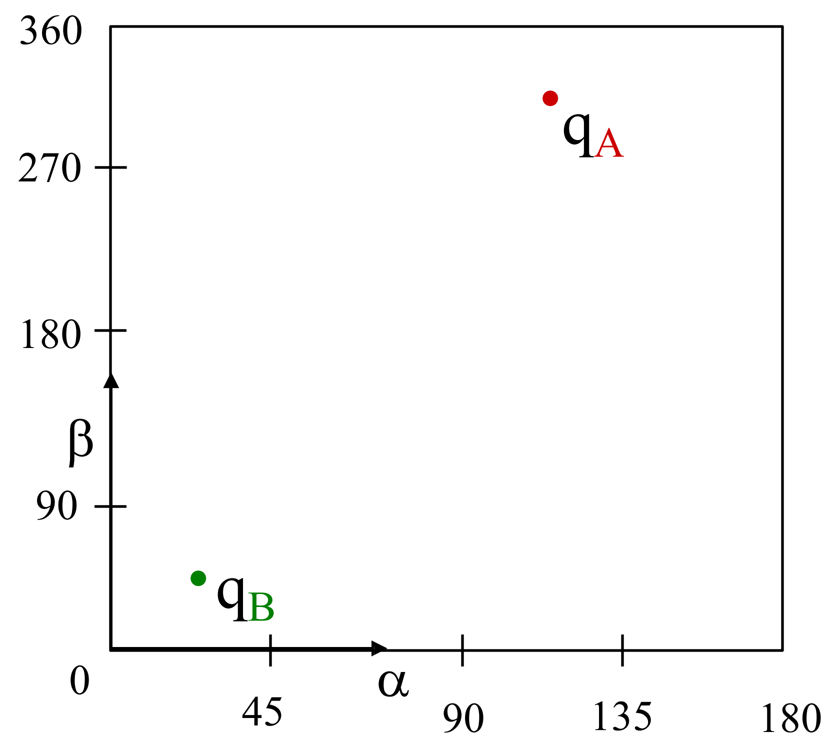
(wraps horizontally and vertically)

Configuration Space "Quiz"

Where do we put  ?



An obstacle in the robot's workspace



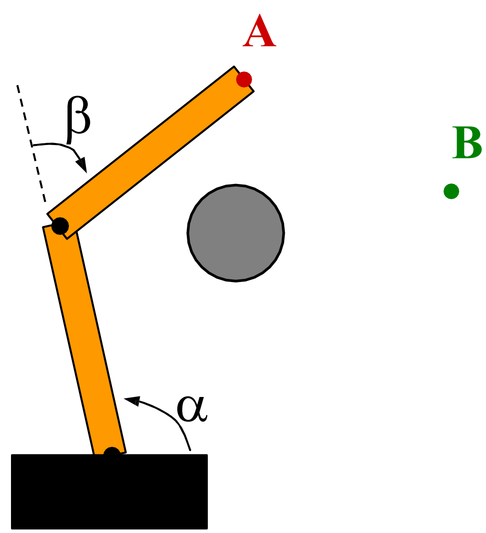
Torus

(wraps horizontally and vertically)

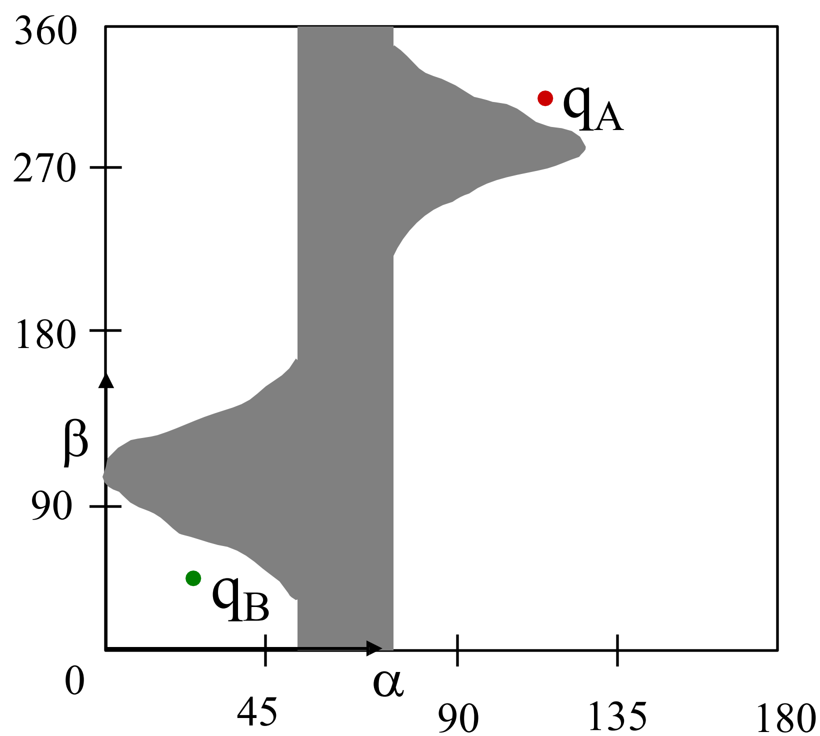
Configuration Space Obstacle

Reference *configuration*

How do we get from **A** to **B** ?

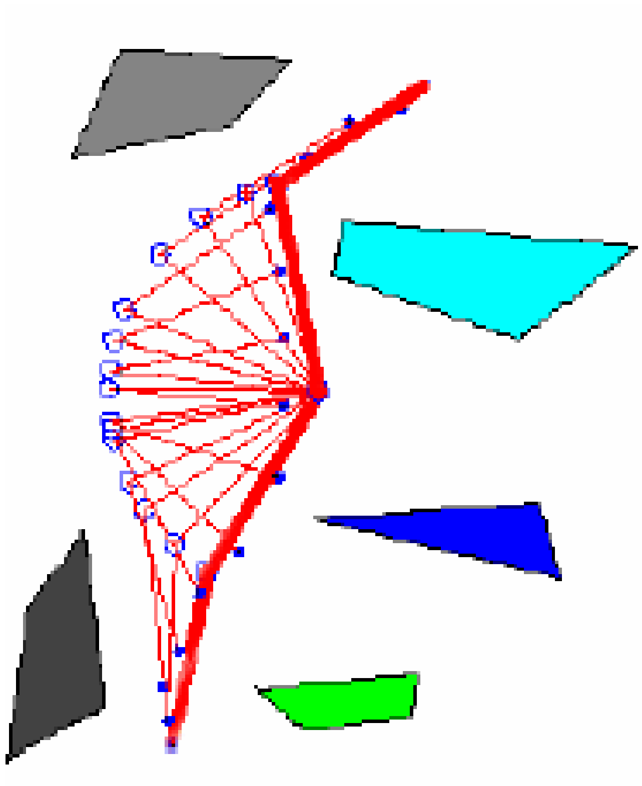


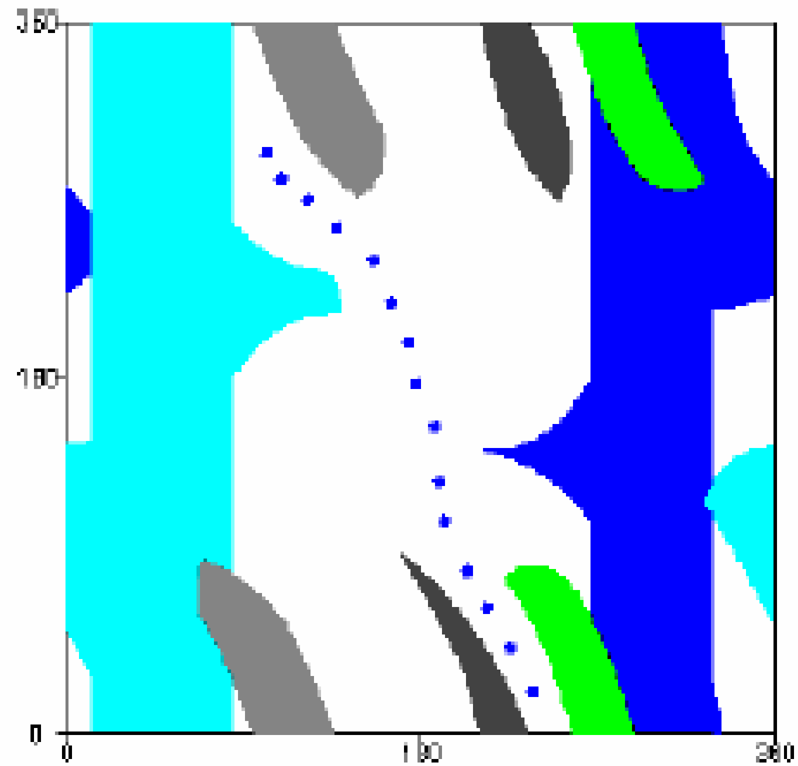
An obstacle in the robot's workspace



The C-space representation of this obstacle…

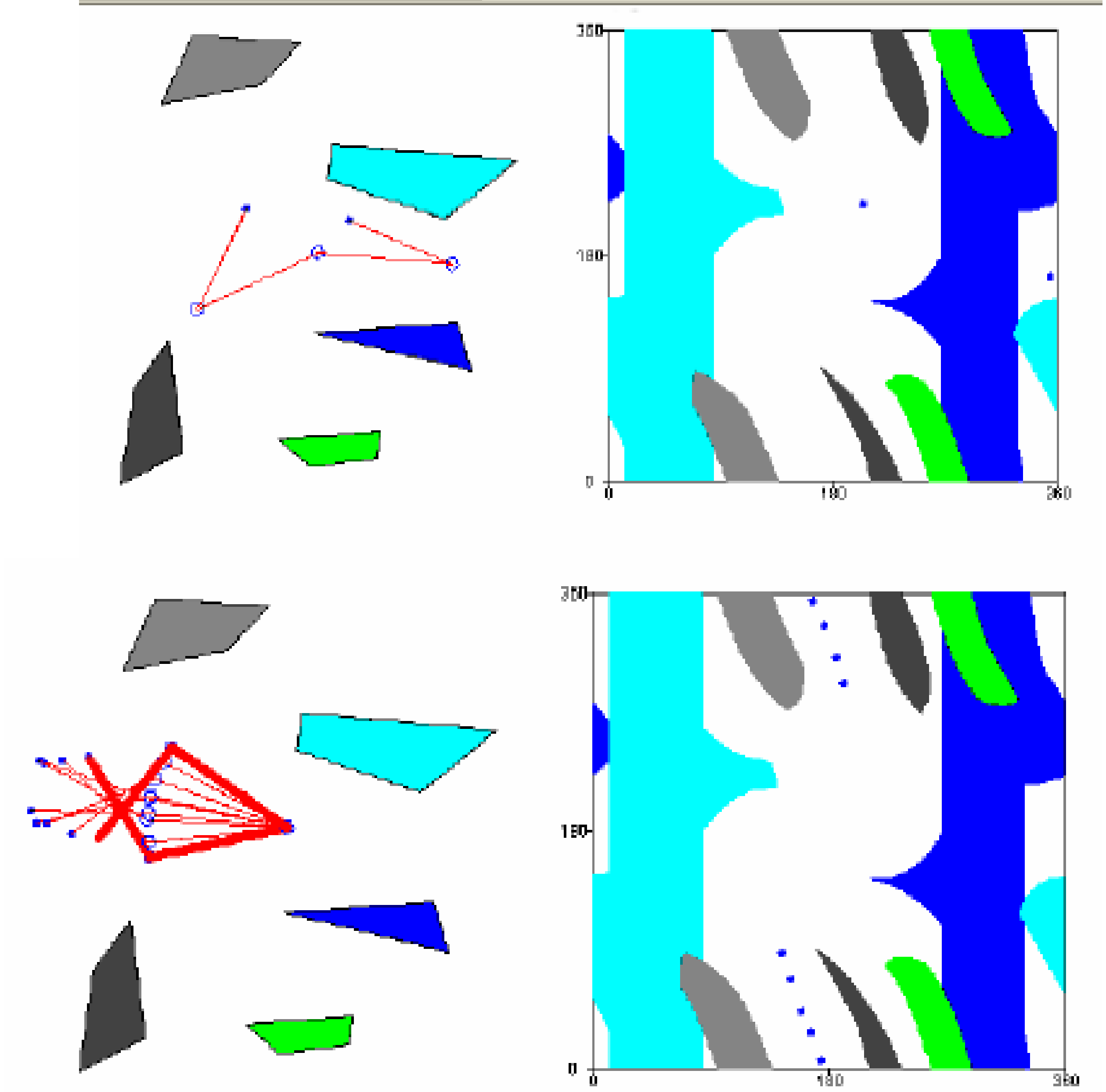
Two Link Path





Thanks to Ken Goldberg

Two Link Path



Properties of Obstacles in C-Space

- If the robot and *WO_i are___________, then*
  - *Convex then QO_i* is convex
  - *Closed then QO_i* is closed
  - *Compact then QO_i* is compact (=closed and bounded)
  - *Algebraic then QO_i* is algebraic
  - *Connected then QO_i* is connected

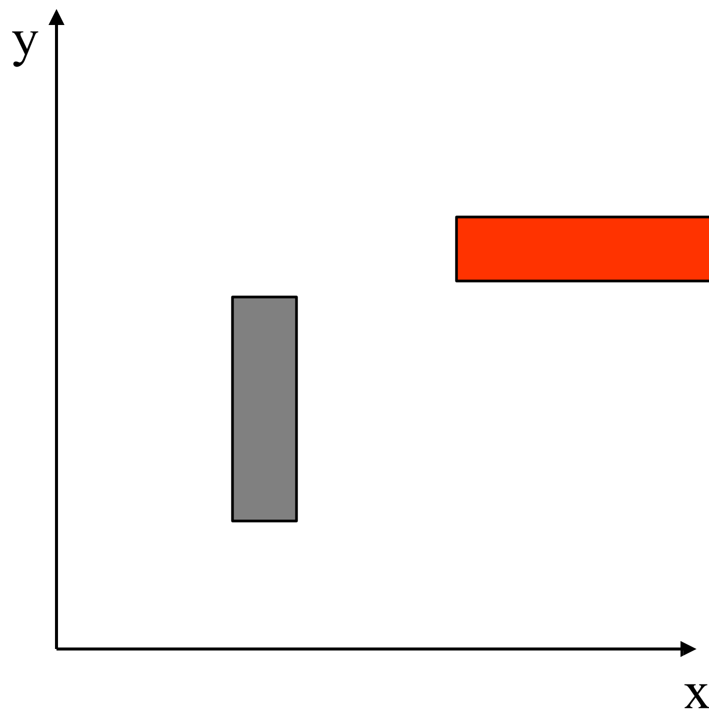
Properties of Obstacles in C-Space

- If the robot and WO_i *are*____________*, then*
  - *Convex then* QO_i is convex ?
  - *Closed then* QO_i is closed ?
  - *Compact then* QO_i is compact (=closed and bounded)
    - *[0, 1] is compact*
    - *[0, 1) is not compact (bounded but not closed)*
    - R^n *is not compact (closed but not bounded)*
    - *Attention:* R^n *is closed <u>and</u> open*
  - *Algebraic then* QO_i is algebraic ? for ex. x2 + y2 – 1 = 0
  - *Connected then* QO_i is connected ?

Additional dimensions

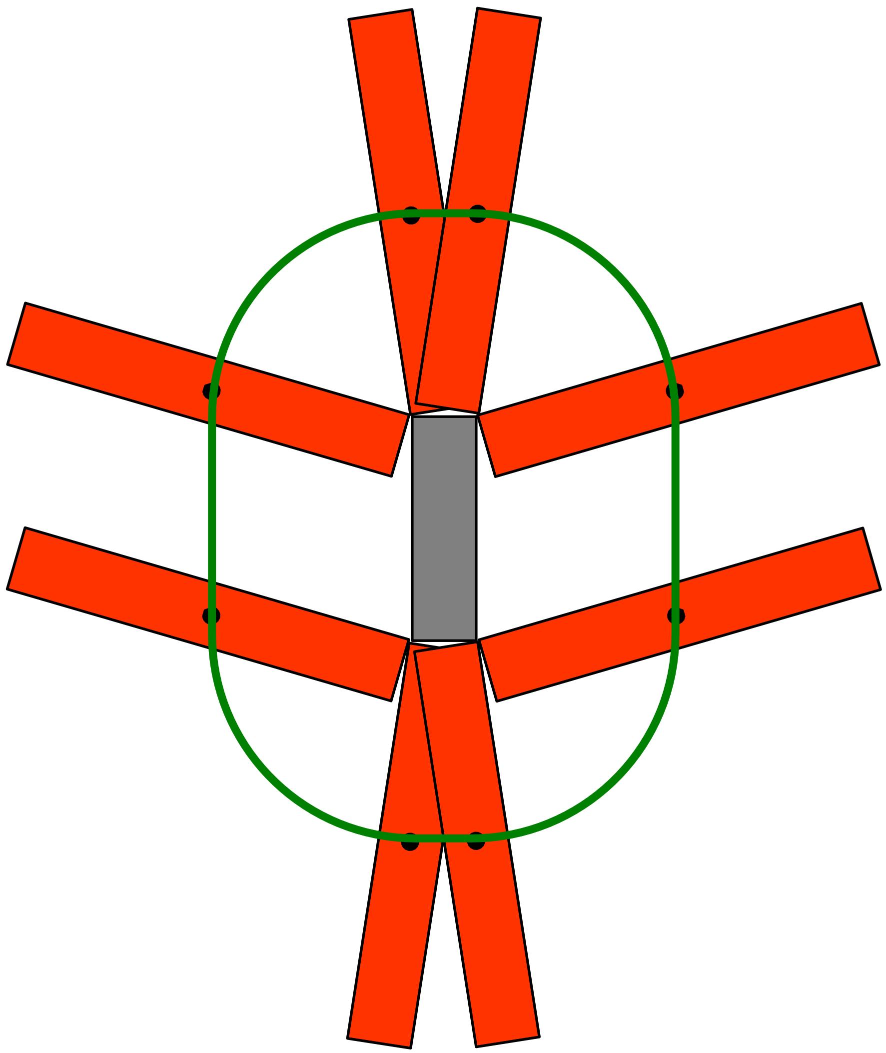
What would the configuration space of a rectangular robot (red) in this world look like? Assume it can translate **and** rotate in the plane.

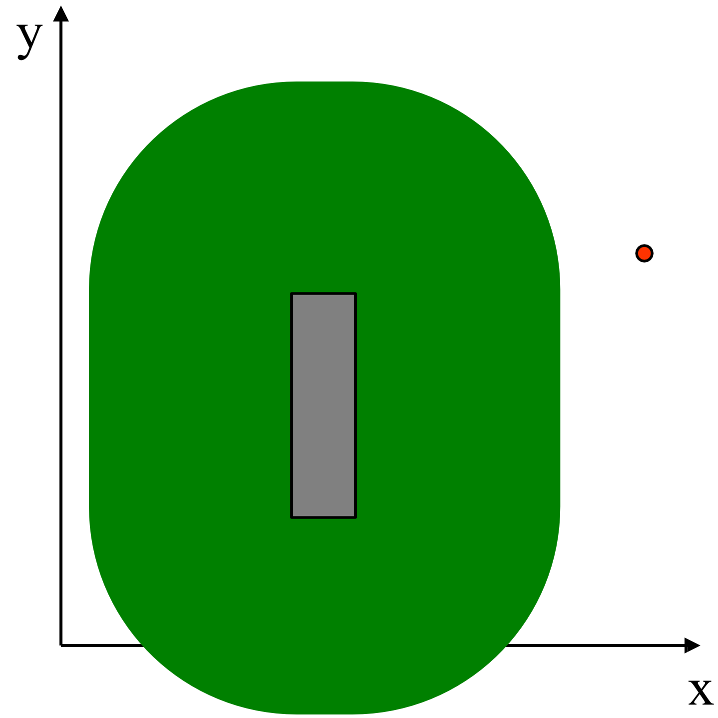
(The grey rectangle is an obstacle.)



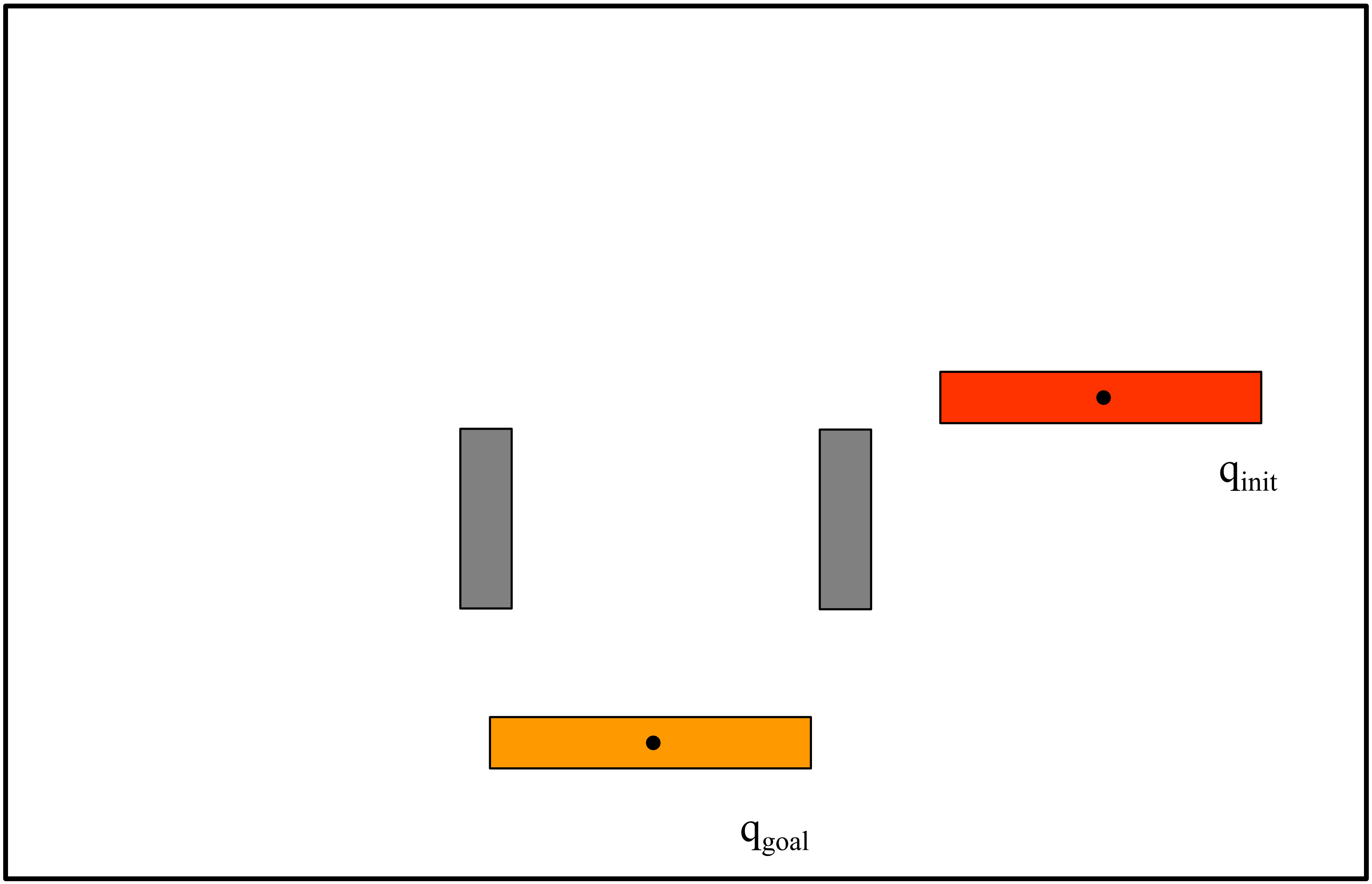
a 2nd possibility

2d projection...



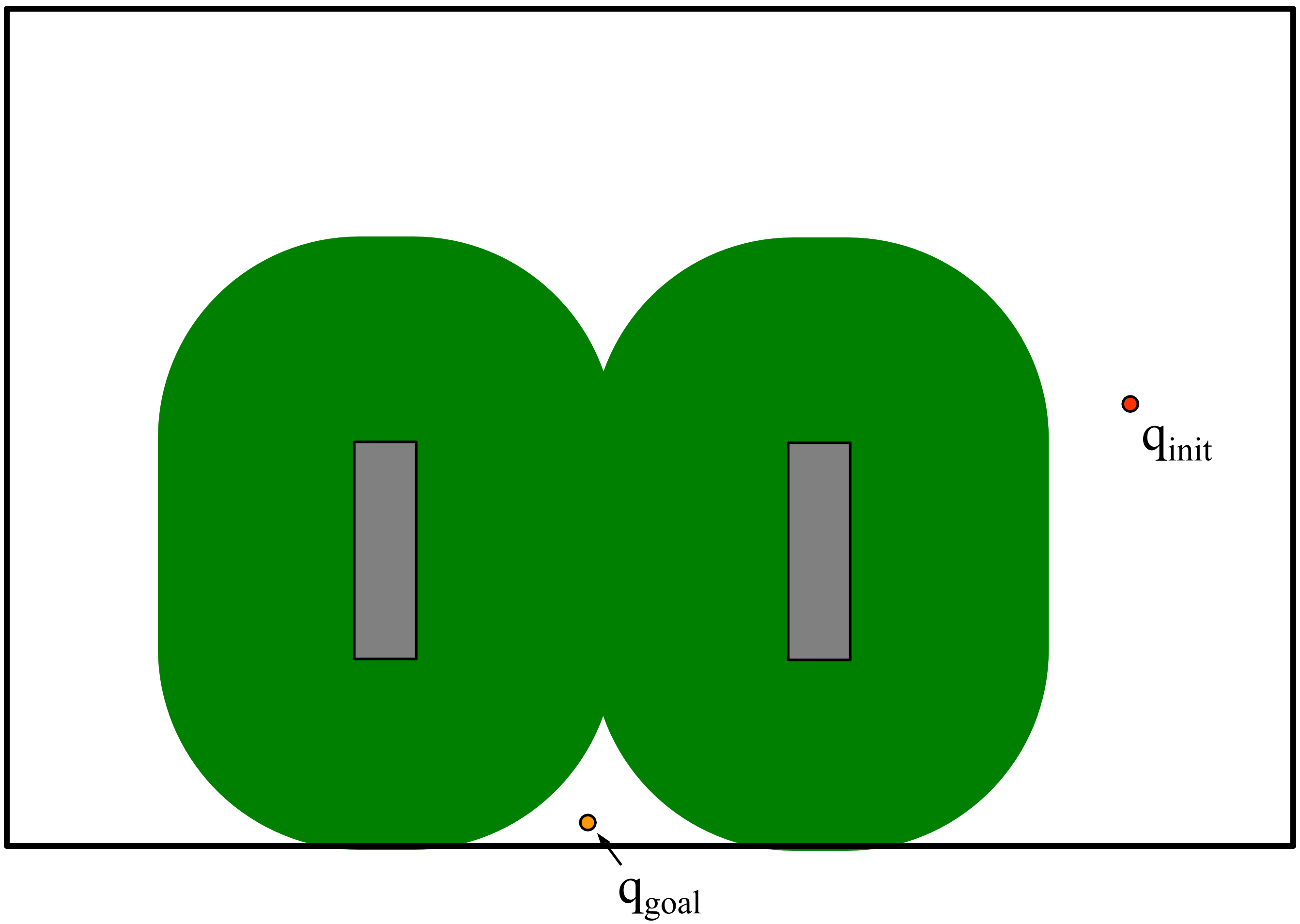


A problem?



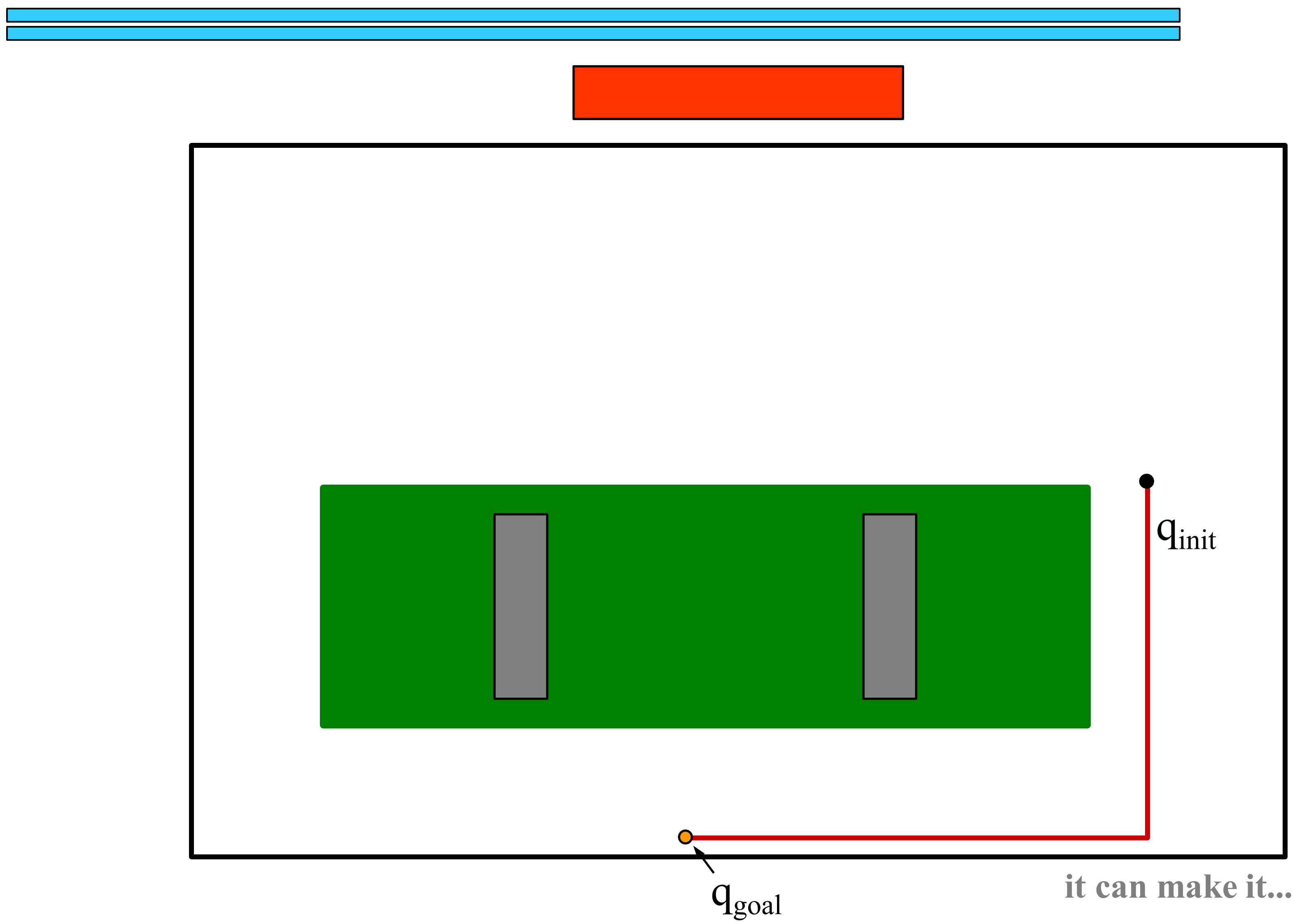
https://math.berkeley.edu/~sethian/2006/Applets/Robotics/java_files_robotic_legal/robotic_legal.html

Requires one more d…



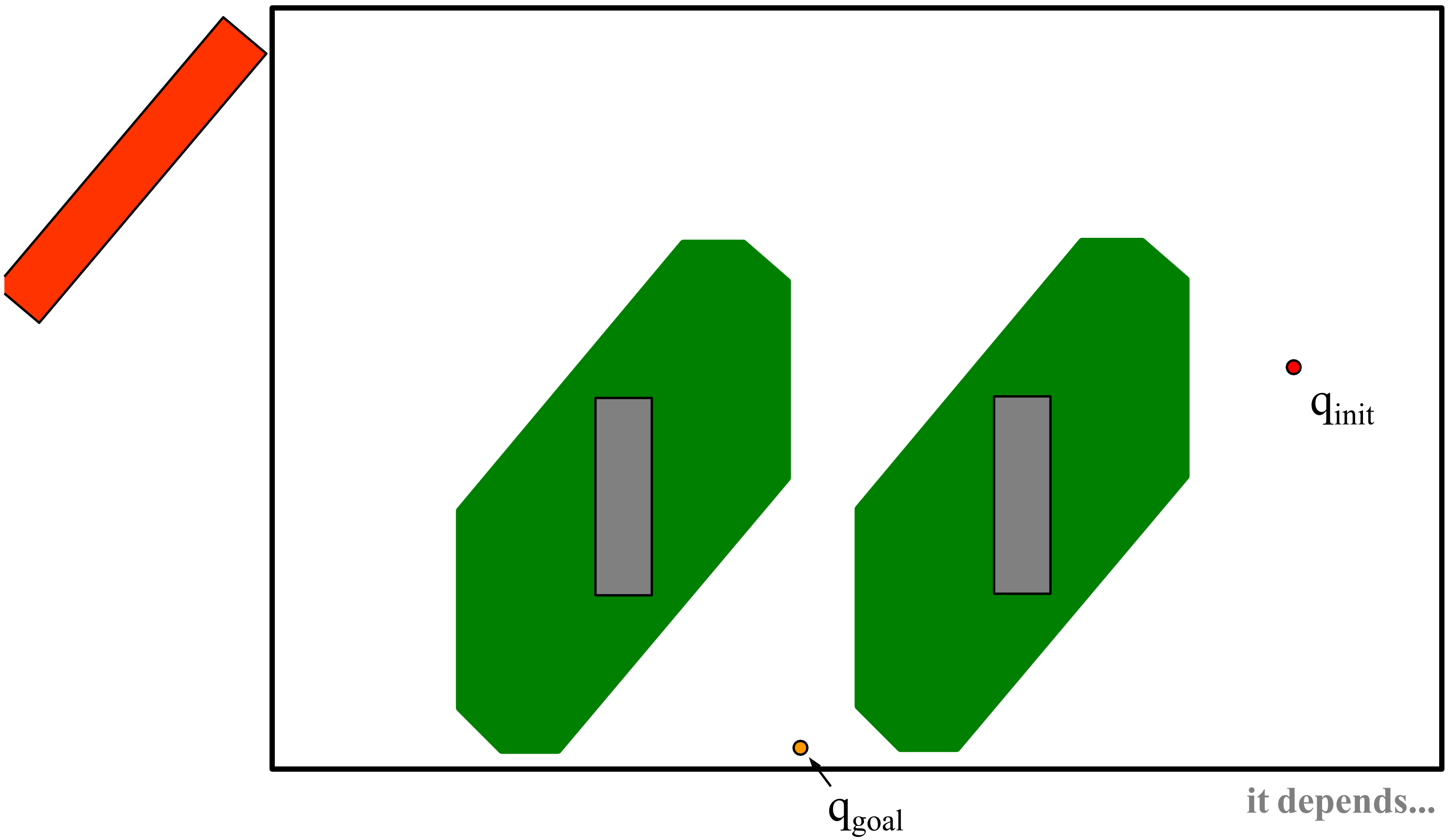
too conservative !
what instead?

When the robot is at one orientation



0°

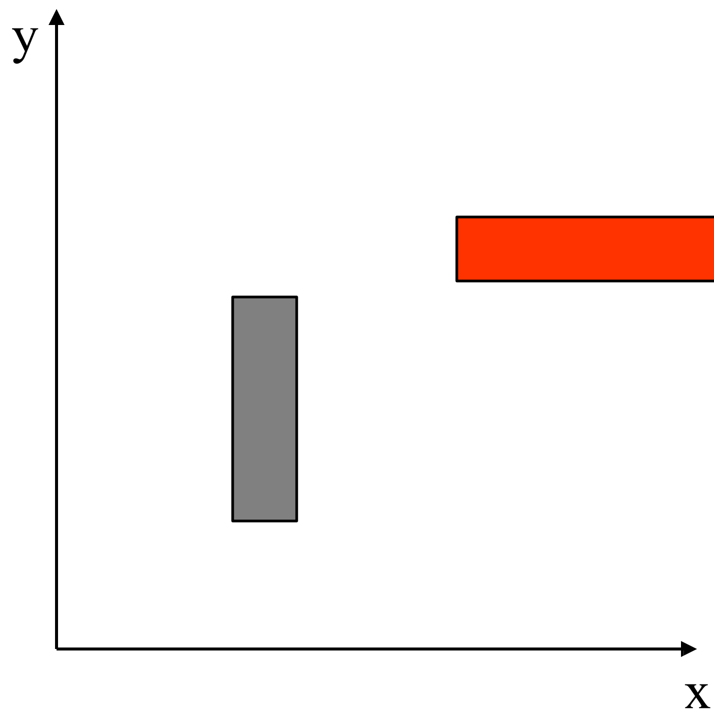
When the robot is at another orientation

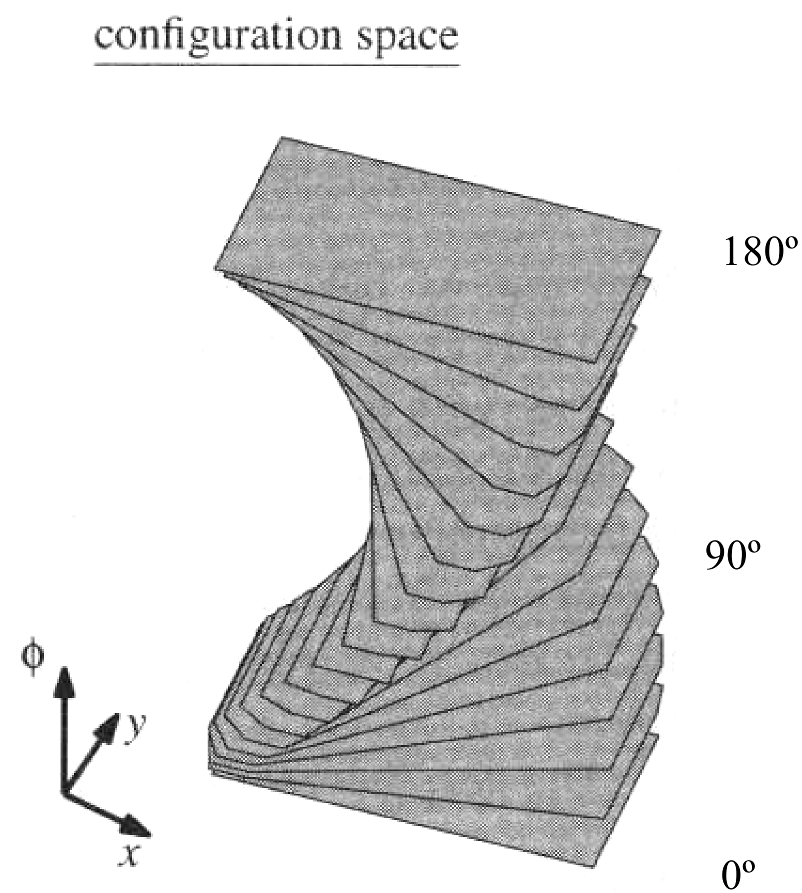


Additional dimensions

What would the configuration space of a rectangular robot (red) in this world look like?

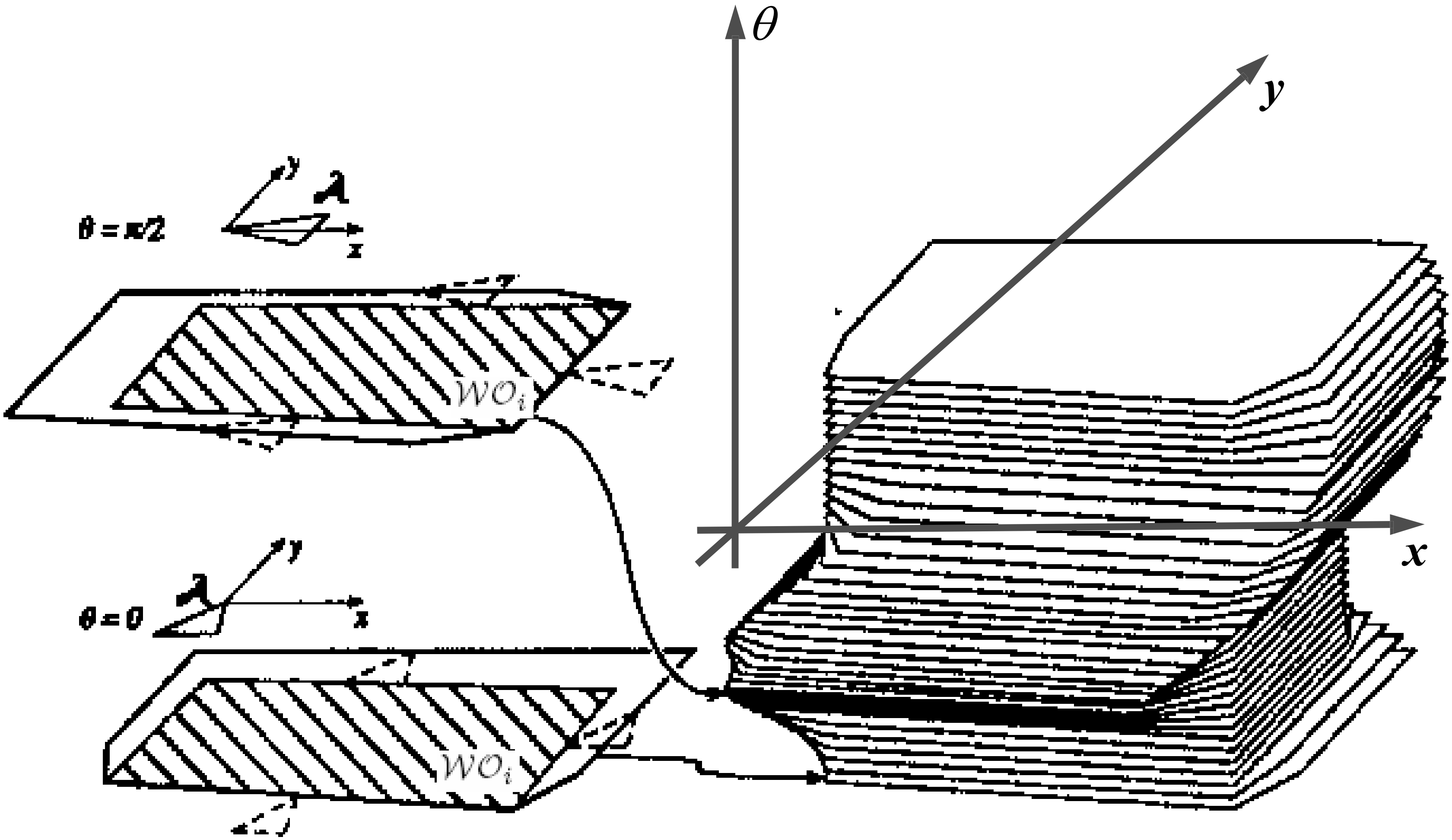
(The obstacle is grey.)



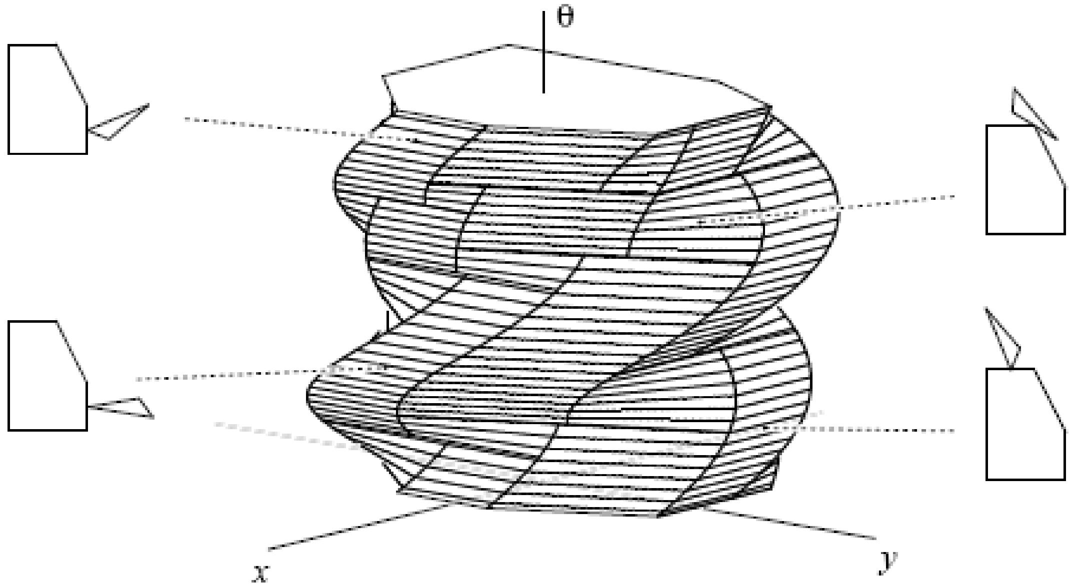


this is twisted...

Polygonal robot translating & rotating in 2-D workspace



SE(2)



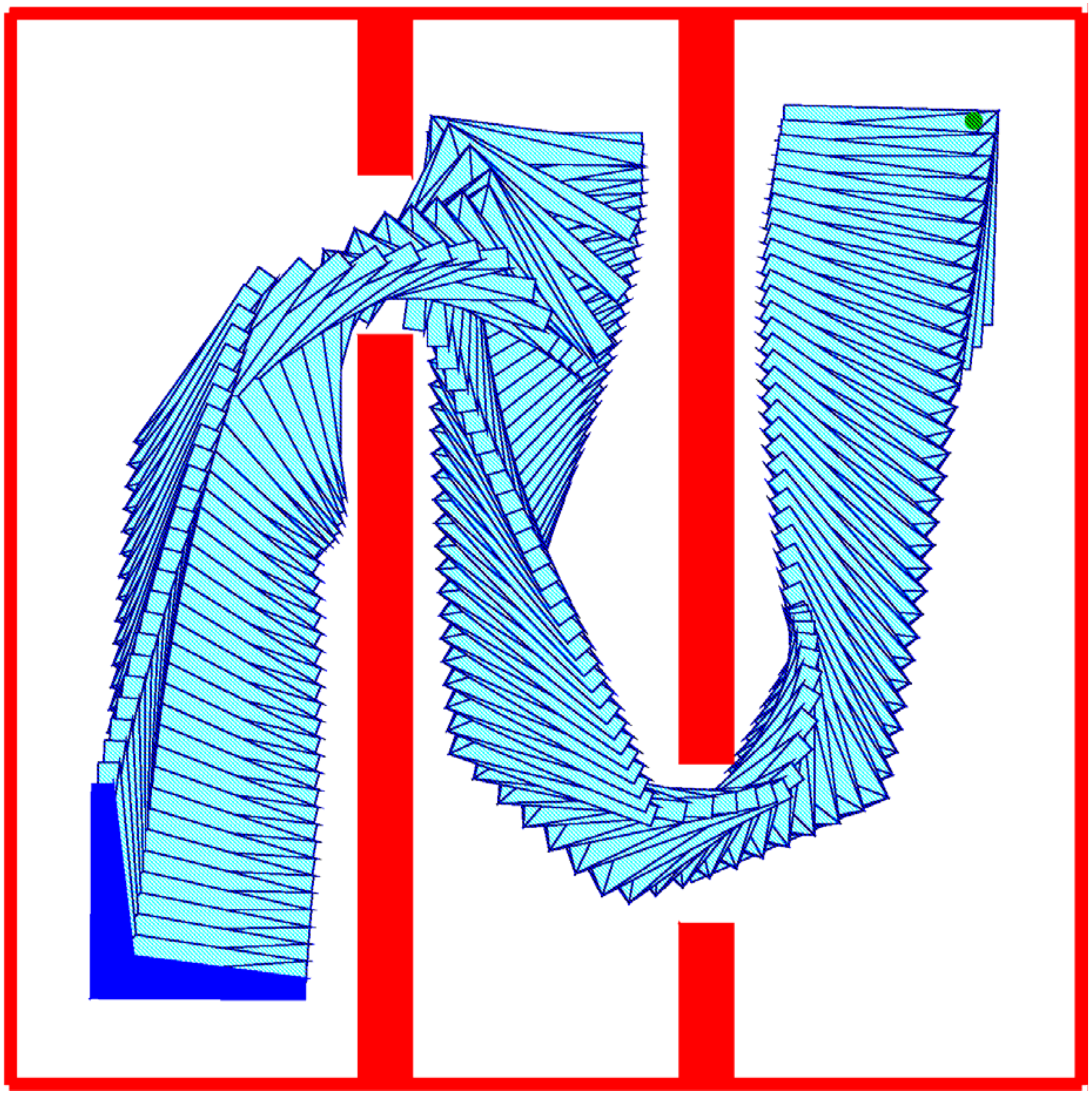
- Special Euclidean group in two dimensions SE(2)

- The set of all 3×3 matrices with the structure:

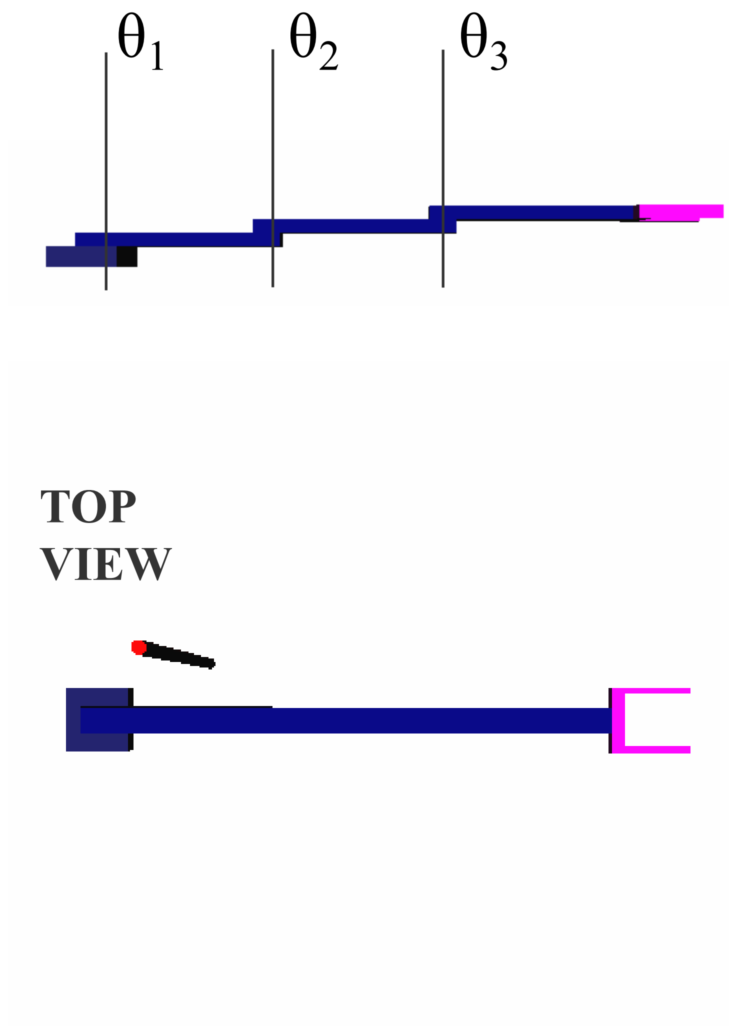
$$\begin{bmatrix} \cos\theta & \sin\theta & x \\ -\sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix}$$

- All planar displacements (translation along x,y and rotation with θ)

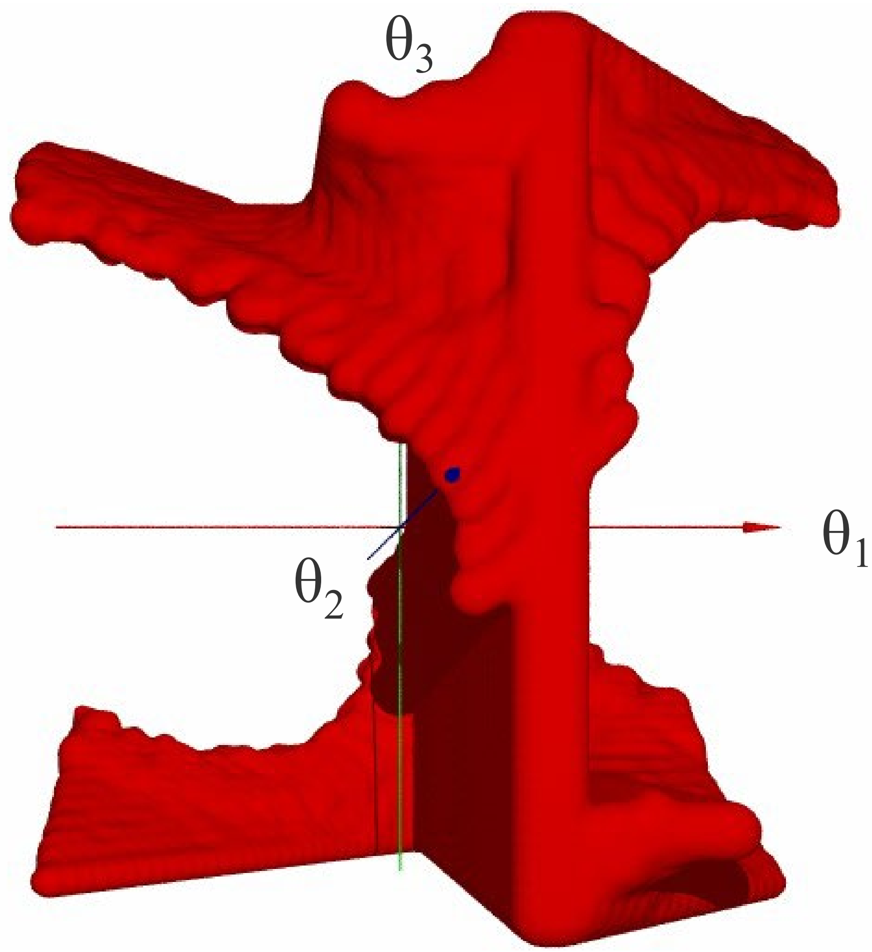
2D Rigid Object



The Configuration Space (*C*-space)

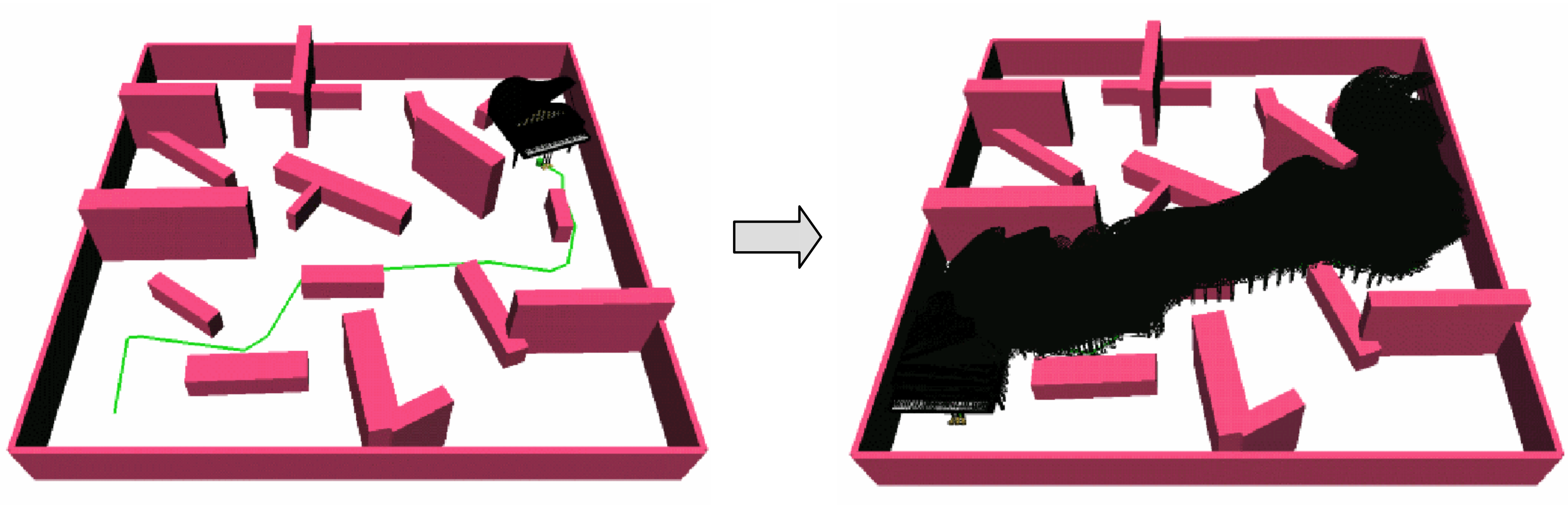


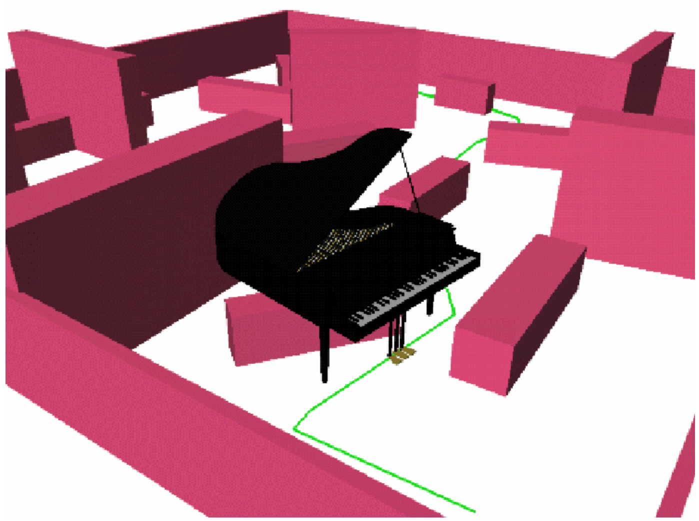
workspace



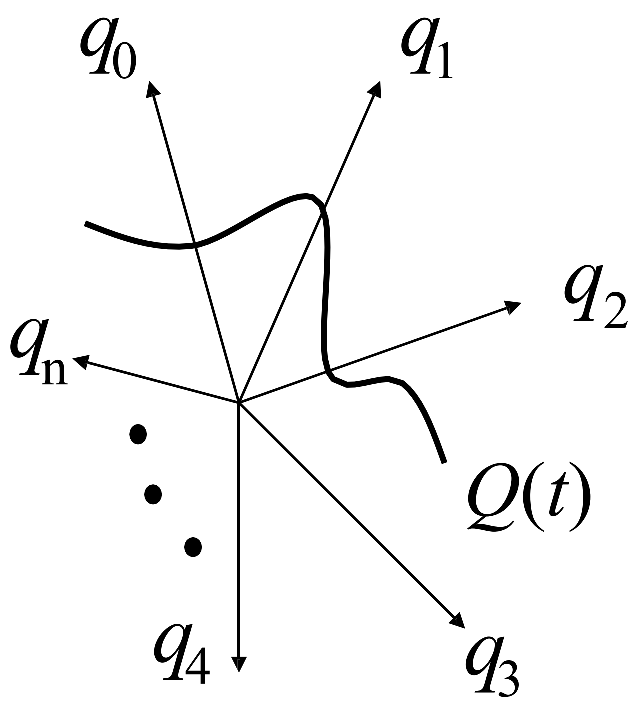
C-space

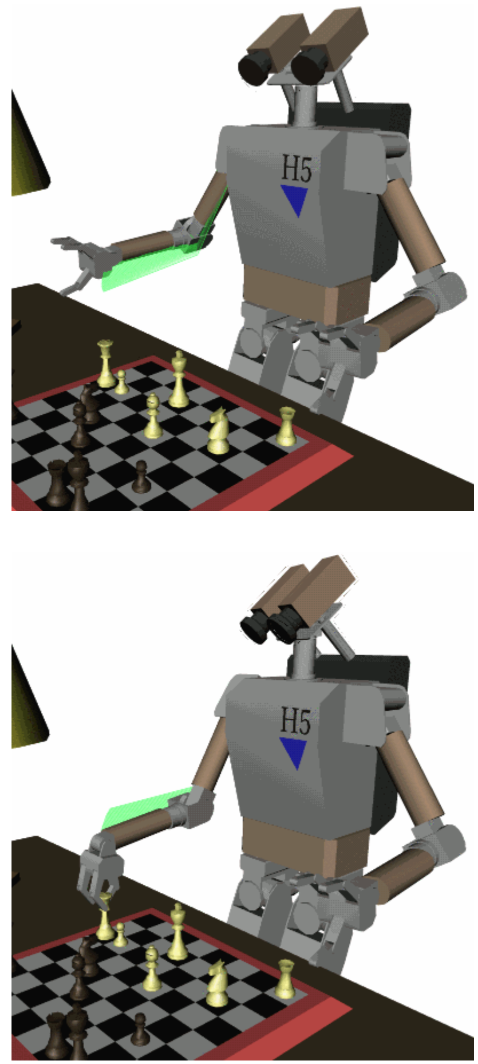
Moving a Piano





Configuration Space (C-space)



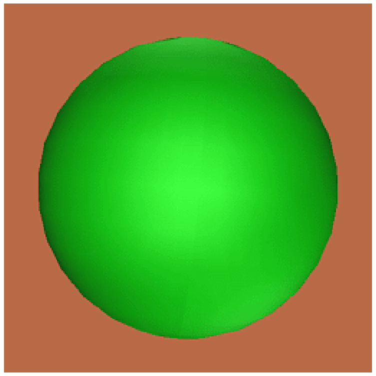


INIT:
$Q(0)$

GOAL:
$Q(T)$

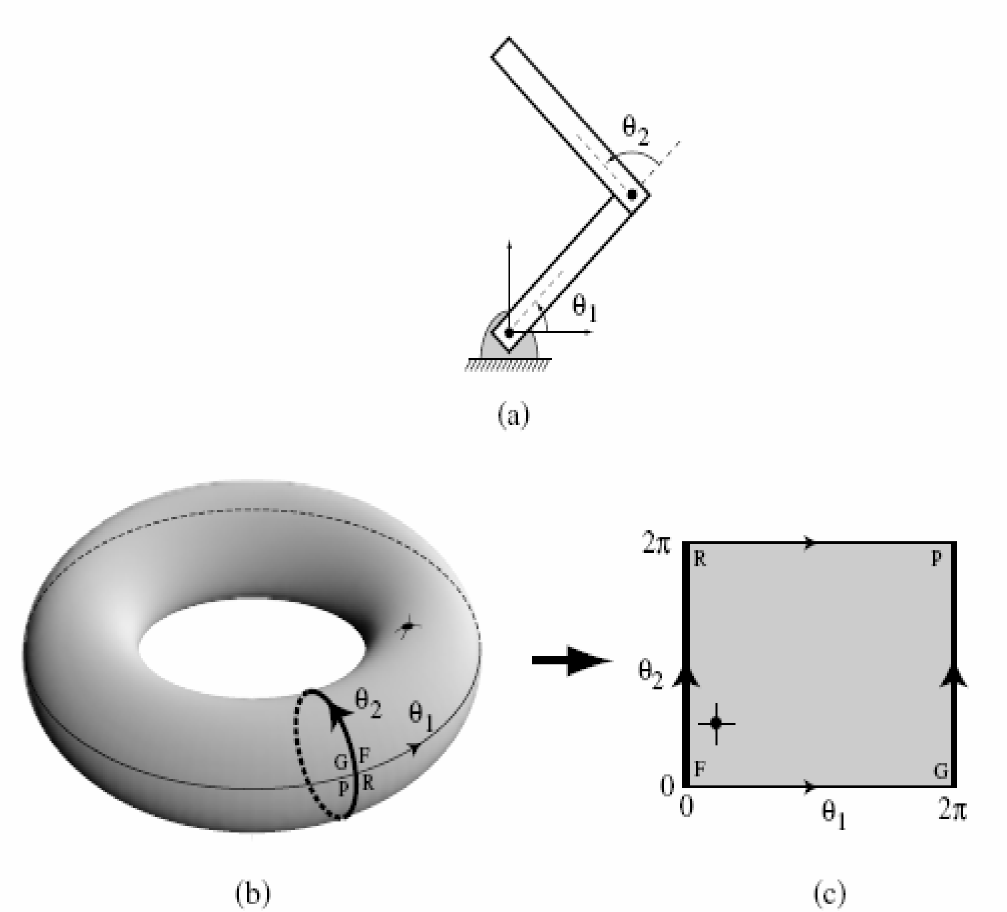
$$Q(t) = \begin{bmatrix} q_0(t) \\ \vdots \\ q_n(t) \end{bmatrix} \quad t \in [0, T]$$

Topology?



Sphere?

Torus?



2R manipulator

Configuration space

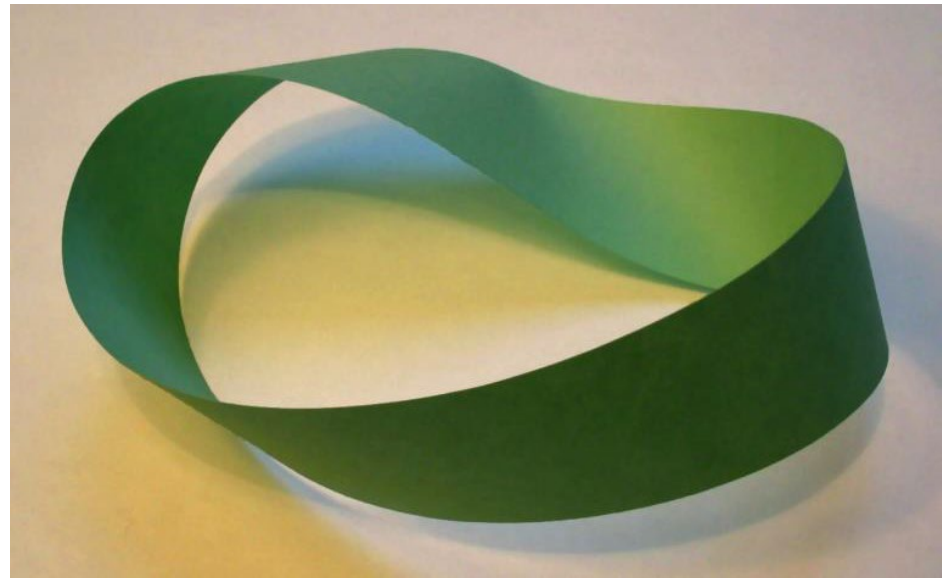
Why study the Topology

- Topology is concerned with
  - the properties of space that are preserved under continuous deformations, such as stretching and bending, but not tearing or gluing.
  - This can be studied by considering a collection of subsets, called open sets, that satisfy certain properties, turning the given set into what is known as a topological space.
  - Important topological properties include connectedness and compactness

- Extend results from one space to another: spheres to stars (navigation function)

The Topology of Configuration Space

- Topology is the "intrinsic character" of a space
- **Two spaces have a different topology if cutting and pasting is required to make them the same (e.g. a sheet of paper vs. a Mobius strip)**
  - think of rubber figures
  - if we can stretch and reshape "continuously" without tearing, one into the other, they have the same topology
- A basic mathematical mechanism for talking about topology is the homeomorphism.



Moebius strip

Sheet of paper

Examples

Type of robot	Representation of Q
Mobile robot translating in the plane	$\mathbb{R}^2$
Mobile robot translating and rotating in the plane	$SE(2)$ or $\mathbb{R}^2 \times S^1$
Rigid body translating in the three-space	$\mathbb{R}^3$
A spacecraft	$SE(3)$ or $\mathbb{R}^3 \times SO(3)$
An n-joint revolute arm	T^n
A planar mobile robot with an attached n-joint arm	$SE(2) \times T^n$

$S^1 \times S^1 \times \ldots \times S^1$ (n times) $= T^n$, the n-dimensional torus

$S^1 \times S^1 \times \ldots \times S^1$ (n times) $\neq S^n$, the n-dimensional sphere in $\mathbb{R}^{n+1}$

$S^1 \times S^1 \times S^1 \neq SO(3)$

$SE(2) \neq \mathbb{R}^3$

$SE(3) \neq \mathbb{R}^6$

More Example Configuration Spaces (contrasted with workspace)

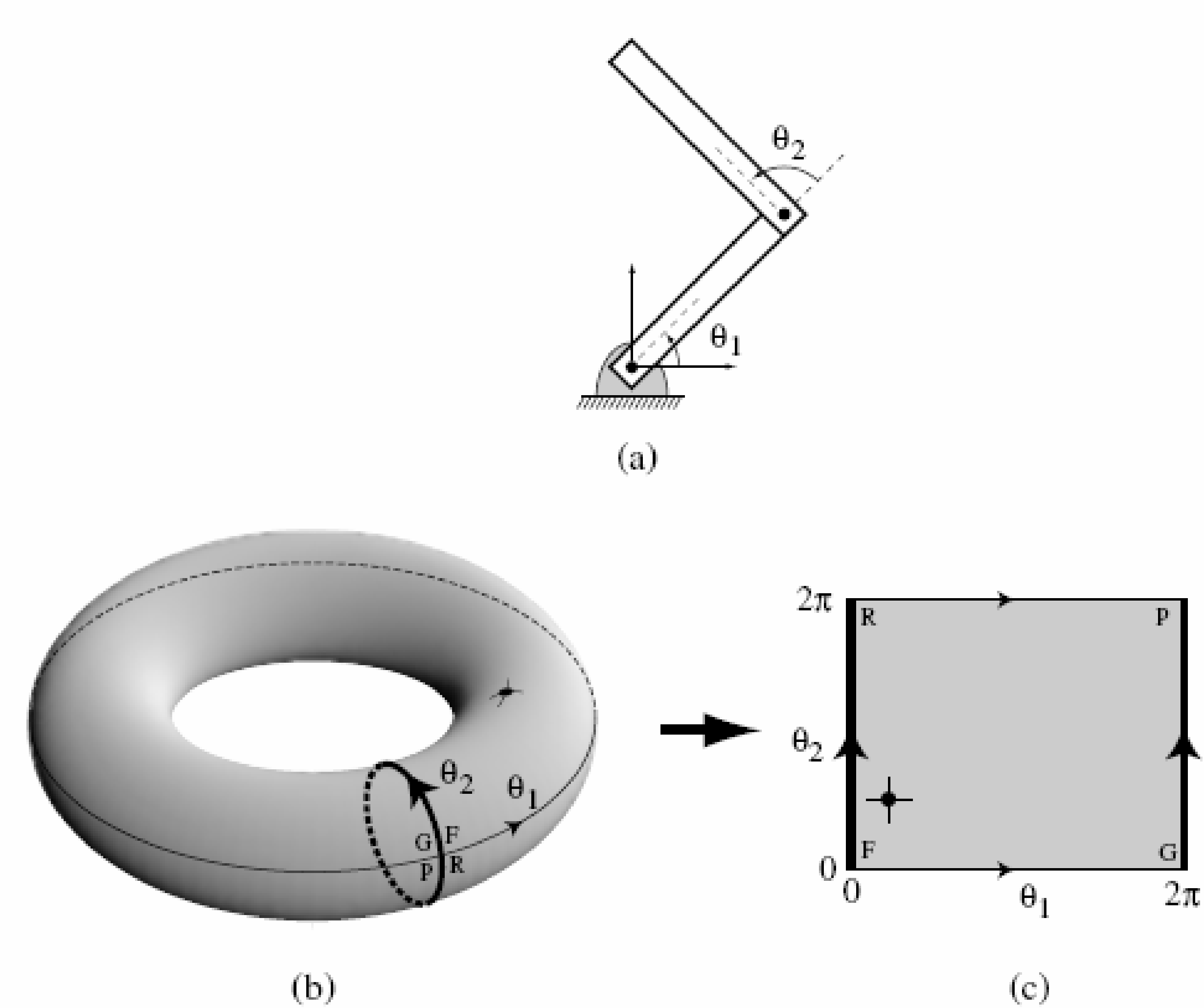
- A rotating bar fixed at a point
  - what is its C-space?
  - what is its workspace (the set of points it can reach?)

- A rotating bar that translates along the rotation axis
  - what is its C-space?
  - what is its workspace

- A two link manipulator
  - what is its C-space?
  - what is its workspace?

- Suppose there are joint limits, does this change the C-space? The workspace?

Parameterization of Torus



$(\theta_1, \theta_2) \in \mathbb{R}^2$

problems at $\theta_i = \{0, 2\pi\}$

Representing Rotations

- Consider S^1 : rotation in the plane
- The action of a rotation is to, well, rotate -> $R_\theta: \Re^2 \rightarrow \Re^2$
- We can represent this action by a matrix R that is applied (through matrix multiplication) to points in $\Re^2$

$$\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

- Note, we can either think of rotating a point through an angle, or rotate the **coordinate system (or frame)** of the point.

Geometric Transforms

Now, using the idea of homogeneous transforms, we can write:

$$p' = \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} \mathrm{p} \qquad \textit{R is 2x2} \quad \textit{t is 2x1}$$

The group of rigid body rotations SO(2) $\times$ $\Re$(2) is denoted SE(2) (for special Euclidean group)

$$R = \begin{bmatrix} \widetilde{x_1} & \widetilde{y_1} \\ \widetilde{x_2} & \widetilde{y_2} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \in SO(2)$$

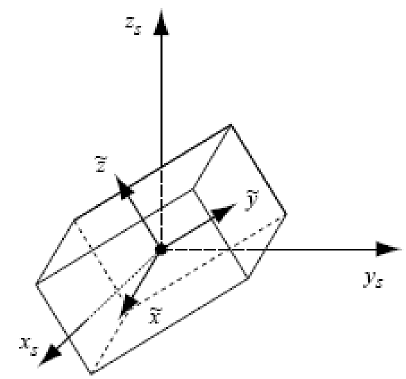
This space is a type of torus

From 2D to 3D Rotation

- I can think of a 3D rotation as a rotation about different axes:
  - rot(x,θ) rot(y,θ) rot(z,θ)
  - there are many conventions for these (see Appendix E)
    - Euler angles (ZYZ) --- where is the singularity
    - Roll Pitch Yaw (ZYX)
    - Angle axis, Quaternion

- The space of rotation matrices has its own special name: SO(n) (for special orthogonal group of dimension n). It is a manifold of dimension n

$$R = \begin{bmatrix} \widetilde{x_1} & \widetilde{y_1} & \widetilde{z_1} \\ \widetilde{x_2} & \widetilde{y_2} & \widetilde{z_2} \\ \widetilde{x_3} & \widetilde{y_3} & \widetilde{z_3} \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \in SO(3)$$



Geometric Transforms

Now, using the idea of homogeneous transforms,
we can write:

$$p' = \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} \mathrm{p} \qquad \textit{R is 3x3} \quad \textit{t is 3x1}$$

The group of rigid body rotations SO(3) $\times$ $\Re$(3) is
denoted SE(3) (for special Euclidean group)

$$SE(n) \equiv \begin{bmatrix} SO(n) & \mathbb{R}^n \\ 0 & 1 \end{bmatrix}$$

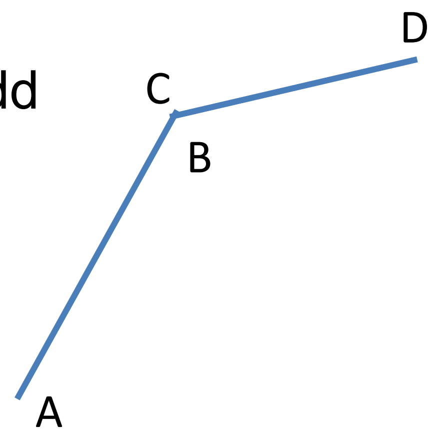
What does the inverse transformation look like?

What is the Dimension of Configuration Space?

- The dimension is the number of parameters necessary to uniquely specify the configuration

- One way to do this is to explicitly generate a parameterization (e.g with our 2-bar linkage)

- Another is to start with too many parameters and add (independent) constraints
  - Now, require $||A-B|| = c_1$ and $||C-D|| = c_2$
  - Now, require B = C
  - Now, fix A = 0



- QUIZ:
  - HOW MANY DOF DO YOU NEED TO MOVE FREELY IN 3-space? How can this be derived from a homogenous transformation?

Summary

- Configuration spaces, workspaces, and some basic ideas about topology
- Kinematics and inverse kinematics
- Coordinate frames and coordinate transformations
- Jacobians and velocity relationships

T. Lozano-Pérez.
Spatial planning: A configuration space approach.
IEEE Transactions on Computing, C-32(2):108-120, 1983.